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Palaeomagnetism of Precambrian dyke swarms in the North China Shield: The ~1.8 Ga LIP event and crustal consolidation in late Palaeoproterozoic times

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ABSTRACT

The North China Shield (NCS) is cut by a laterally-extensive dyke swarm emplaced at 1.78–1.76 Ga when an extensional regime succeeded regional metamorphism and completion of cratonisation by ~1.85 Ga. Palaeomagnetic study of these dykes and adjoining metamorphic country rocks identifies a dominant shallow axis comprising a contiguous population with NE to N declinations and rare opposite polarity. Dykes with NE shallow magnetic declination (A1, $D/I = 36/-1°$) recognised from previous study and emplaced in granulite terranes in the north are displaced by more northerly declinations (A2, $D/I = 8/2°$) in lower grade metamorphic terranes to the south. Contact tests indicate a primary cooling-related origin to these magnetisations although tests are in part ambiguous because magnetisations in the granulite basement are comparable. Petrologic and rock magnetic considerations imply that magnetisation of the dykes occurred during uplift from depths as deep as 20 km following the peak of metamorphism at ~1.85 Ga. A temporal migration A2 → A1 is implied by the higher crustal level and earlier acquisition of the former, and the deeper source and later acquisition of the latter. A third population of dyke magnetisations (A3, $D/I = 18/43°$) is distributed towards steeper inclinations and close to the Mesozoic-Recent palaeofield. These are either partial or complete overprints of A1–A2 magnetisations with greater degrees of alteration indicated by demagnetisation and thermomagenetic spectra, or are much younger dykes of Mesozoic-Tertiary age. A minority fourth (later Precambrian but presently undated) dual polarity population has a magnetisation (11 dykes, $D/I = 108/7°$) with contact tests indicating a primary cooling-related origin.

The ~1.78–1.76 Ga time of emplacement of the dominant dyke swarms in this study is widely represented by contemporaneous igneous rocks in other major shields linked to major Large Igneous Province (LIP)-related events. The new definition of a ~1.83–1.76 Ga APW swathe from the North China Shield permits a comparison with other shields and yields a constraint to continental configurations during the late Palaeoproterozoic. A quasi-integral reconstruction of Palaeopangaea is tested here and supported by conformity of predominantly of uplift-related palaeopoles from the ~1.90–1.70 Ga tectono-thermal belts and from SW → NE trending APW implied by the distribution of poles from the ~1.80 Ga igneous suites including the LIP events. This trend incorporates the A2 → A1 migration and the granulite terrane cooling polar swathe from North China. The reconstruction indicates that continental crust consolidated in Palaeoproterozoic times by accretion of ~2.3–1.7 Ga orogenic belts around a hemispheric and crescent-shape core already established by Late Archaean times. The North China Shield is interpreted to have bordered the western cratonic margin of the Indian Shield in a proximity supported by correlation of geological features and suggested by a number of previous workers. The Central Orogenic Zone of the North China shield characterised by tectono-thermal activity prior to ~1.85 Ga was then contiguous with a comparable zone running through the centre of the Indian Shield and continuing into the Capricorn Belt of Western Australia. The ~1.78–1.76 Ga dykes in North China continue into dyke swarms in the South India Shield and may have been sourced in a plume-related LIP focussed near the continental margin in the Xiong'er Aulacogen.

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1. Introduction

Laterally-extensive mafic dyke swarms are a key feature of the Precambrian shields and are a signature of large igneous provinces (LIPs) which have been widely linked to up-rise of mantle plumes beneath the continental crust. It is becoming increasingly apparent that they were also emplaced during a relatively small number of concentrated episodes (Yale and Carpenter, 1998; Abbott and Isley, 2002). Thus long dyke swarms ~1.80 Ga in age occur in several shields, as do swarms ~1.25 Ga in age with the latter well exposed as the Mackenzie–Sudbury swarms spanning the Laurentian Shield and the Jotnian activity in the Fennoscandian Shield. Although often interpreted as signatures of continental break-up, this was not necessarily the case; the Mackenzie (1.267 Ga) and Sudbury (1.235 Ga) swarms for example, comprise outcrop radiating across the entire width of the Laurentian Shield from a focus near Coppermine Bay in the North West Territories and have never been the site of subsequent continental break-up. The later Franklin episode (0.723 Ga) is another swarm that does not appear to parallel the locus of any subsequent continental break-up.

In this paper we report palaeomagnetic study of a dense swarm of predominantly northerly-trending dykes cutting across the central part of the North China Shield (NCS). This swarm is well constrained to the interval ~1.78–1.76 Ga by precision dating and correlates temporally with major swarms in a number of other Precambrian terranes. The palaeomagnetic results, together with previous studies of this dyke swarm, collectively define apparent polar wander (APW) for the shield during these times and enable the configuration of the NCS to be resolved with respect to other continental shields late in Palaeoproterozoic times.

2. Geological framework

The crust of the NCS is largely of Archaean antiquity with the most important phase of growth being 2.7–2.5 Ga (Cui et al., 1989; Li et al., 1997; Guan et al., 2002; Kusky and Li, 2003; Kusky et al., 2007b). It comprises three major subdivisions, namely the Eastern Block (EB) and Western Block (WB) (see Fig. 1) separated by a broad SSW–NNE trending zone influenced by Palaeoproterozoic tectonism. The WB is a composite zone comprising the Yinshan Block in the north welded to the Ordos Block in the south along a ~1.95–1.92 Ga collisional belt characterised by khondalite (sillimanite-garnet gneisses) and forming the Inner Mongolia Suture (Fig. 1, inset, Zhao et al., 2000; Kusky et al., 2007c; Santosh et al., 2010). The Ordos Block was welded to the EB by one or more collisional events before ~1.85 Ga (Kusky et al., 2001; Santosh et al., 2010). The Central Zone between them (CZ, Fig. 1 and see Zhao et al., 2000; Zhai and Liu, 2003), otherwise referred to as the Trans-North China (or Central) Orogen, records the signature of a number of tectonic events (Kusky et al., 2007c; Santosh et al., 2010) commencing before 2.6–2.5 Ga (Kröner et al., 1998); it also includes remnants of ~2.5 Ga ocean crust (Kusky et al., 2004). Metamorphism and subsequent magmatism in this zone have been interpreted either as the signature of up-welling of a mantle plume (Zhai and Liu, 2003), or as the consequence of continental collision that welded the EB and WB (Zhao et al., 2000, 2002a,b) with geochemical affinities to magmatic arc and intra-arc environments tending to favour the latter model (Wang et al., 2004). The time of welding of the EB and WB along the CZ is currently disputed with some workers favouring an early ~2.5 Ga age (e.g. Kusky et al., 2007b,c) and others a late ~1.85 Ga age (e.g. Kröner et al., 2005).

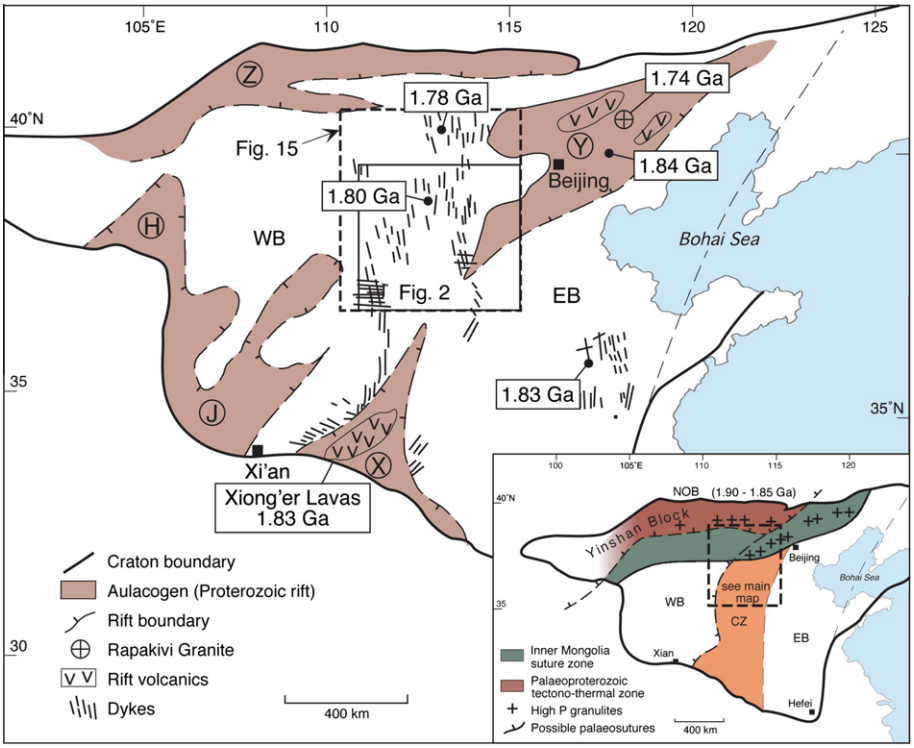


Fig. 1. Outline map of the North China Shield showing the distribution of Palaeoproterozoic dyke swarms. Also shown is the distribution of Palaeo-Mesoproterozoic age aulacogens. Some high precision age determinations linked to the ~1800 Ma LIP event are located and the approximate distribution of the 1.83 Ga Xiong'er volcanics is shown. The aulacogens are abbreviated as J, Jinshan, X, Xiong'er, Y, Yanliao and Z, Zaertai-Bayan Obo. The two enclosed blocks on this figure show the extent of Fig. 2 embracing the present palaeomagnetic study with the larger block including the distribution of palaeomagnetic studies of the North China dykes considered in the Discussion, listed in Table 3 and shown in Fig. 14. The inset figure shows the major tectonic divisions of the North China Shield abbreviated as: WB, Western Block, CZ, Central Zone, EB, Eastern Block and NOB, Northern Orogenic Belt.

Although Palaeoproterozoic dyke emplacement spans the entire width of the NCS, members are best exposed within the CZ where they are emplaced into high grade granulite-grade terranes in the north (e.g. Zhao et al., 2004a,b,c; Kröner et al., 2005; Peng et al., 2005, 2007), decreasing southwards into amphibolite and greenschist facies in the Wutai-Fuping terrane (Liu et al., 2002; Kusky et al., 2007a). High pressure assemblages comprise a belt more than 700 km in length including the Inner-Mongolia-North Hebei Belt (~2.1–1.9 Ga) where P–T–t trajectories record crustal thickening followed by uplift and orogenic collapse along low angle extensional shear zones (Qian et al., 1985; Wilde et al., 1997; Lu, 1991; Lu et al., 2002; Kröner et al., 2005; Zhang et al., 2007). Regional granulite facies metamorphism peaked at 1.88–1.85 Ga; subsequently 1.85–1.80 Ga east–west left-lateral shear zones segmented the shield into belts of Palaeoproterozoic reworking separated by Archaean nuclei (Kusky et al., 2007c) and are delineated by the aeromagnetic signature (Zhang et al., 2007).

Cratonisation of the NCS was completed following regional metamorphism at ~1.85 Ga and welding of the EB and WB along the CZ. Subsequent tectonism was dominated by extension and rift systems developed mainly from the margins of the NCS and mafic dyke swarms of great lateral continuity were emplaced; these are most abundant within the central part of the shield. Five aulacogens which developed within the NCS broadly contemporaneous with dyke emplacement comprise the Xiong'er, Jinshan, Yanliao, Zhaertai-Bayan Obo and Helanshan rifts (Fig. 1). The NNE-SSW trending Xiong'er–Zhongtiao rift at the southern extremity includes lavas of the Xiong'er Group dated between 1.8 and 1.76 Ga (Zhao et al., 2004a,b,c; He et al., 2009, Zhao et al., 2010) and mafic dykes yielding 1.80 to 1.40 Ga U–Pb zircon ages (Sun and Hu, 1993; Lu et al., 2002 and Zhao et al., 2004a,b,c). The Xiong'er is interpreted as the signature of an LIP by Peng et al. (2005, 2007) with the dykes probably recording lava feeders and illustrating a fanning geometry focussed within the Xiong'er rift (Peng et al., 2006). Other aulacogens recording the Late Palaeo-Mesoproterozoic history of the NCS and shown in Fig. 1 include the Jinshan (Shanxi-Shanxi) now truncated in the south by the late Proterozoic North Qinling orogenic belt, and the Yanliao-Taihang incorporating sediments and volcanics of the 1.85–1.40 Ga Changcheng System deposited on basement up to 500–700 Ma older (Kusky et al., 2007c). Sediments of the Jixian (~1.6–1.4 Ga) and Qingbaikou (~1.0–0.8 Ga) systems record subsequent deposition on the Fuping Craton. The latter incorporates rapakivi granites dated 1706 ± 31 and 1697 ± 2 Ma (Yu et al., 1994) with contemporaneous anorogenic anorthosites in North Hebei dated between 1730 ± 16 and 1693 ± 7 Ma (Zhao et al., 2004a,b,c). The Zhaertai-Bayan Obo is margin-parallel and infilled with 1.80–1.40 Ga sedimentary and volcanic successions (Kusky et al., 2007c). The Helanshan aulacogen in the west has a sedimentary infill deposited on Archaean-early Proterozoic basement although the age is not yet well constrained.

Swarms comprising several hundred N-NW trending dykes with a minority of E–W members are widely exposed within the NCS and most notably within the CZ. They include the Taihang tholeiite swarm over 20 km wide and 300 km long reaching from Taihang Mountain to southern Inner Mongolia, which together with related

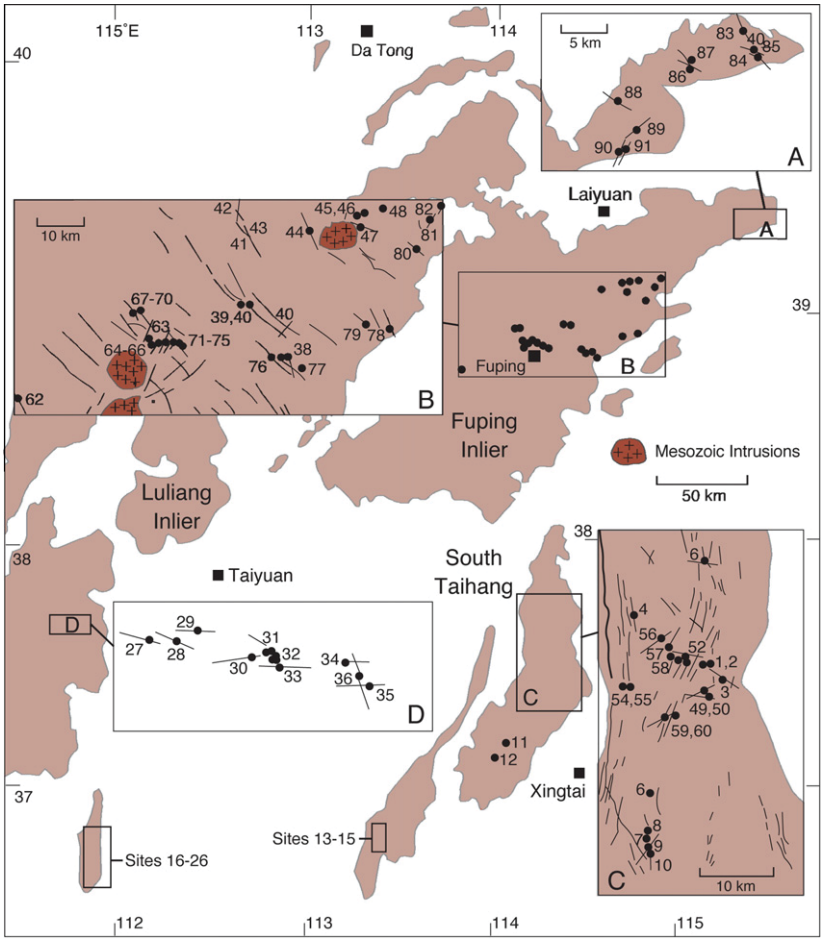


Fig. 2. Regional distribution of the dykes sampled for this study. The shaded area is the outcrop of Archaean and Early Proterozoic crust and the location of this region within the North China Shield is shown as the solid box in Fig. 1.

swarms, comprises an igneous episode extending for a strike length of at least 1000 km across the NCS (Qian and Chen 1987; Hou et al., 2001, 2006; Kusky et al., 2007c; Peng et al., 2004, 2005, 2007). Most dykes are basic in composition but a few are acidic. Peng et al. (2007) recognise two generations of emplacement with an earlier low Fe–Ti group emplaced at ~1780 Ma and a younger high Fe–Ti tholeiitic group emplaced at ~1760 Ma. They are vertical to subvertical in orientation and chilled margins record sharp contacts with the country rocks; they contrast with suites of deformed dykes metamorphosed up to granulite facies which are a facet of the pre-1.85 Ga history (Peng et al., 2005, 2007). The NW swarm locally cuts the lowermost formations in the Yanliao aulacogen (Fig. 1) and is cross-cut by more quartz-rich tholeiites of a WNW swarm (Hou and Mu, 1994; Halls et al., 2000).

The early age of many of these dykes is evidenced by their unconformable setting beneath the Changcheng Group considered to be older than ~1.70 Ga (Wang et al., 1997, 2007; Peng et al., 2008) and with a 1761 ± 16 Ma age possibly representative of the higher part of this succession (Zhao et al., 2002a,b); they are also likely to be genetically linked to the Xiong'er Group with a younger age of ~1.75 Ga. Peng et al. (2005) obtained a SHRIMP zircon U–Pb age of 1778 ± 3 Ma and zircon/baddeleyite $^{207}Pb/^{206}Pb$ age of 1777 ± 26 Ma for a mafic dyke at Datong in the Western Block, and U–Pb ages of 1754 ± 71 Ma for a NNW dyke and 1789 ± 28 Ma for an E–W dyke. A dolerite dyke at Mt. Hengshan in the Central Orogenic Belt has yielded a single-grain zircon U-Pb age of 1769.1 ± 2.5 Ma (Halls et al., 2000) and Wang et al. (2004) report $^{40}Ar/^{39}Ar$ plateau ages of 1780.7 ± 0.5, 1765 ± 1.1 and 1774.7 ± 0.7 Ma from the Taiyuang Inlier (see Fig. 2). In western Shandong zircon age determinations from NNW-trending mafic dykes in the range 1844–1790 Ma (Hou et al., 2006; Shao et al., 2005) may include inherited material and are probably too high; other Rb–Sr, Nd–Sm and K–Ar age data defining less precise ages of emplacement are discussed by Peng et al. (2008). The NCS has been subject to dyke emplacement during several later times including the Neoproterozoic and Mesozoic-Tertiary (Shao et al. (2005). Sometimes the distinction between swarms of different ages is apparent from field evidence; this is often however, not the case and in these circumstances the palaeomagnetic and rock magnetic results from the dykes provide key criteria for assigning dykes to their respective age groups (Sections 4 and 5).

Subsequent segmentation of the shield by faulting has produced the present topography of rugged inliers in faulted or unconformable contact with Proterozoic, Phanerozoic and Mesozoic strata infilling intervening depocentres. The NE-trending extensional basins are delimited by normal faults defining uplifted boundaries to the inliers and possible planes of rotation (Fig. 3). Faulting and block tilting occurred during the Mesozoic and Cenozoic following closure of the PalaeoTethys Ocean during tectonism that correlates broadly with collision of the NCS with the Asian continental collage and subsequent strike slip motions.

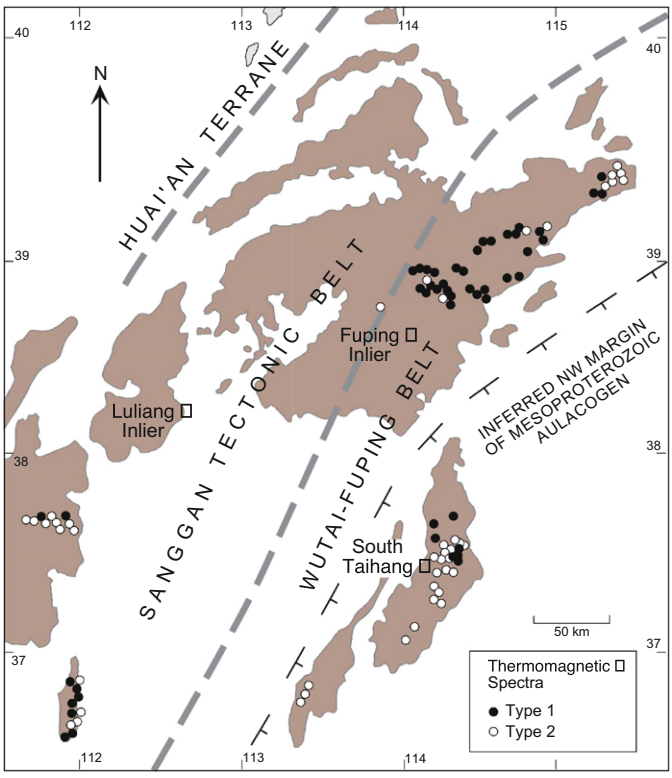


Fig. 3. Distribution of major tectonic divisions of the study area with the distribution of Type A and Type B thermomagnetic spectra determinations.

3. Field and laboratory methods

Cores for palaeomagnetic and rock magnetic study were collected using a portable field drill or as oriented blocks from 91 sites at locations summarised in Table 1. The bulk of the sampled units are dykes although 10 sites were drilled in adjoining metamorphic basement rocks of amphibolite to granulite metamorphic grade at, or near to, dyke contacts. Cores were cut into cylinders initially for measurement of magnetic susceptibility, and followed by measurement of magnetic remanence using mostly 'Minispin' magnetometers or otherwise nitrogen SQUID (FIT) magnetometer for weakly magnetised samples. Thermal demagnetisation was conducted in steps of 50 °C to 500 °C and then in reduced steps of 20 °C until directional behaviour ceased to be systematic. Components comprising the Natural Remanent Magnetisation (NRM) were resolved by eye from orthogonal projections of the remanence vector onto horizontal and vertical planes and equivalent directions calculated by Principal Component Analysis (PCA). Rock magnetic tests included thermomagnetic determinations performed on at least one piece from each site using a computer controlled horizontal translation Curie balance; investigations of hysteresis and Isothermal Remanent Magnetisation (IRM) acquisition were performed on samples from 37 sites using a variable frequency translation balance (VFTB) or pulse magnetiser.

4. Rock magnetic results

Examples of thermomagnetic spectra (saturation magnetisation, M_s, versus temperature) are illustrated in Fig. 4 with Curie points defined using the method of intercepting tangents (Grommé et al., 1969) and listed in Table 1. Although a number of more complex curve classifications have been proposed for these spectra, a simple twofold scheme is appropriate to this study: a Type 1 comprises convex curves yielding discrete titanomagnetite Curie points close to pure magnetite in composition (Fig. 4); these tend to be stable to heating and yield RM values close to 1 (Table 1), although mostly somewhat less than this, presumably due to oxidation of magnetite to hematite during the heating and cooling in air.

Type 1 sites are dominant in the NW sector of the study area and characterise the central part of the Fuping Craton. The second group (Type 2) is composite: convex signatures of magnetite are either a subsidiary signature or absent and a prominent concave asymptotic portion is the signature of paramagnetism obeying the Curie Law; higher parts of the curves may identify discrete Curie points near to pure hematite (~680 °C), although this tends to be less clearly recognisable because hematite has a saturation remanence only 1–2% of magnetite. Some Type 2 curves are stable to

Table 1
Summary of Palaeomagnetic site locations, rock types and thermomagnetic properties.

Site no.	Locality	Rock type	Orientation	Longitude	Latitude	Curie point(s)	
							Type, RM
1	Zanhuang	M	215/55	114.19.40	37.32.53	660	**2**, 0.88
2	Zanhuang	Dyke	40/68	114.17.16	37.32.33	680	**2**, 1.03
3	Zanhuang	Dyke	35/78	114.17.16	37.32.33	–	**2**, 1.14
4	Zanhuang	M	265/70	114.9.48	37.35.10	570, 680	**1**, 0.89
5	Zanhuang	Dyke	133/85	114.13.44	37.34.27	565, 670	**1**, 0.90
6	Zanhuang	Dyke	180/88	114.12.4	37.27.7	680	**2**, 0.84
7	Xingtai	Dyke	320/32	114.11.36	37.24.54	580	**2**, 0.89
8	Xingtai	Dyke	180/85	114.11.36	37.24.54	570	**2**, 0.91
9	Xingtai	Dyke	305/48	114.11.44	37.24.48	640	**1**, 0.90
10	Xingtai	Dyke	95/80	114.11.44	37.25.48	685	**2**, 1.11
11	Xingtai	Dyke	300/10	114.8.36	37.4.49	685	**2**, 1.02
12	Zuoquan	M	0/15	114.3.24	37.5.6	585	**2**, 1.07
13	Zuoquan	M	290/81	113.22.40	37.6.18	575	**2**, 1.00
14	Zuoquan	M	307/66	113.22.40	37.6.18	585	**2**, 0.92
15	Zuoquan	M	310/62	113.22.36	37.6.12	585	**2**, 1.09
16	Zuoquan	Dyke	235/87	111.49.48	36.25.36	580	**1**, 1.22
17	Huoxian	Dyke	225/90	111.49.48	36.25.36	585	**1**, 1.73
18	Huoxian	M	38/42	111.49.48	36.25.36	585	**2**, 1.03
19	Huoxian	Dyke	225/90	111.49.36	36.25.36	580	**2**, 2.67
20	Huoxian	Dyke	35/85	111.49.48	36.25.36	585	**2**, 1.00
21	Huoxian	M	40/78	111.49.48	36.25.36	580	**2**, 1.05
22	Huoxian	Dyke	262/80	111.49.48	36.25.36	580	**1**, 0.90
23	Huoxian	Dyke	270/85	111.49.48	36.25.36	585	**1**, 0.80
24	Huoxian	M	108/82	111.55.16	36.36.6	585	**1**, 1.02
25	Huoxian	M	108/82	111.56.4	36.36.8	610	**2**, 1.10
26	Huoxian	M	110/80	111.56.0	36.36.8	585	**1**, 0.88
27	Huoxian	Dyke	205/71	111.41.28	37.39.32	585	**2**, 2.47
28	Jiaocheng	Dyke	202/82	111.42.4	37.39.25	680	**2**, 1.18
29	Jiaocheng	Dyke	181/75	111.42.36	37.39.41	585	**1**, 0.81
30	Jiaocheng	Dyke	178/82	111.43.44	37.39.14	580	**2**, 1.16
31	Jiaocheng	Dyke	190/73	111.44.4	37.39.18	580	**2**, 1.99
32	Jiaocheng	Dyke	200/75	111.44.8	37.39.10	585	**2**, 1.45
33	Jiaocheng	Dyke	180/70	111.44.11	37.39.7	680	**2**, 2.01
34	Jiaocheng	Dyke	178/79	111.45.40	37.39.0	580	**1**, 1.25
35	Jiaocheng	Dyke	185/86	111.46.0	37.38.42	680	**2**, 0.85
36	Jiaocheng	Dyke	70/86	111.46.8	37.38.35	680	**2**, 0.90
37	Jiaocheng	M	310/29	114.24.16	38.50.10	580	**1**, 0.72
38	Jiaocheng	Dyke	220/85	114.24.12	38.50.13	575	**1**, 0.95
39	Fuping	Dyke	228/88	114.19.8	38.55.45	580	**1**, 0.92
40	Fuping	Dyke	65/68	114.19.8	38.55.45	580	**1**, 0.90
41	Fuping	Dyke	248/76	114.20.40	39.2.36	585	**1**, 0.85
42	Fuping	Dyke	280/60	114.21.36	39.3.52	585	**1**, 0.68
43	Fuping	Dyke	250/90	114.21.36	39.3.52	580	**1**, 0.80
44	Fuping	Dyke	250/88	114.28.8	39.6.3	580	**1**, 0.93
45	Laiyuan	Dyke	248/85	114.35.29	39.7.40	585	**1**, 0.98
46	Laiyuan	Dyke	105/90	114.35.29	39.7.40	585	**1**, 0.73
47	Laiyuan	Dyke	210/88	114.35.31	39.6.32	580	**1**, 0.92
48	Laiyuan	Dyke	70/89	114.35.26	39.7.47	585	**2**, 0.78
49	Zanhuang	Dyke	335/35	114.17.39	37.32.20	585	**1**, 0.80
50	Zanhuang	Dyke	195/70	114.17.39	37.32.20	580, 670	**1**, 1.20
51	Zanhuang	Dyke	325/60	114.15.50	37.33.37	590	**1**, 0.94
52	Zanhuang	Dyke	195/90	114.15.50	37.33.52	585	**2**, 1.03
53	Zanhuang	M	258/64	114.15.20	37.33.41	580	**1**, 0.80
54	Zanhuang	Dyke	218/53	114.15.20	37.33.41	660	**2**, 0.63
55	Zanhuang	Dyke	178/90	114.15.20	37.33.41	590, 689	**2**, 0.78
56	Zanhuang	Dyke	145/87	114.13.40	37.34.37	–	**2**, 0.72
57	Zanhuang	M	280/45	114.14.20	37.33.56	340, 630	**2**, 0.56
58	Zanhuang	Dyke	332/57	114.14.41	37.33.46	585	**2**, 1.09
59	Zanhuang	Dyke	160/80	114.13.43	37.30.58	590	**2**, 1.21
60	Zanhuang	Dyke	313/82	114.14.18	37.30.58	–	**2**, 1.18
61	Fuping	Dyke	11/75	114.16.47	37.38.30	585	**1**, 1.82
62	Fuping	Dyke	68/62	113.46.7	38.44.29	585	**2**, 1.82
63	Fuping	Dyke	225/90	114.5.40	38.52.22	630	**2**, 1.42
64	Fuping	Dyke	235/50	114.6.53	38.52.2	580	**1**, 0.91
65	Fuping	Dyke	112/75	114.6.53	38.52.2	580	**1**, 0.88
66	Fuping	Dyke	115/88	114.6.58	38.52.1	580	**1**, 0.85
67	Fuping	Dyke	46/68	114.7.37	38.51.49	585	**1**, 0.82
68	Fuping	Dyke	110/90	114.7.37	38.51.49	580	**1**, 1.08
69	Fuping	Dyke	145/45	114.7.37	38.51.49	580	**1**, 580
70	Fuping	Dyke	43/69	114.7.38	38.51.46	585	**1**, 0.79
71	Fuping	Dyke	106/82	114.8.2	38.51.40	585	**1**, 0.79
72	Fuping	Dyke	108/85	114.8.2	38.51.40	580	**1**, 0.92
73	Fuping	Dyke	110/89	114.8.33	38.51.29	–	**2**, 1.20

Table 1 (*continued*)

Site no.	Locality	Rock type	Orientation	Longitude	Latitude	Curie point(s)	
							Type, RM
74	Fuping	Dyke	128/88	114.8.36	38.51.25	580	**1**, 0.90
75	Fuping	Dyke	60/74	114.9.31	38.51.6	580	**1**, 0.92
76	Fuping	Dyke	42/89	114.24.9	38.50.9	575	**1**, 0.90
77	Fuping	Dyke	55/67	114.26.19	38.48.47	580	**1**, 0.95
78	Fuping	Dyke	78/86	114.40.2	38.54.14	575	**1**,0.90
79	Fuping	Dyke	80/83	114.40.2	38.54.14	575	**1**, 0.85
80	Fuping	Dyke	228/87	114.43.7	39.4.23	570	**2**, 1.55
81	Fuping	Dyke	110/71	114.46.4	39.8.20	585	**1**, 0.96
82	Yixian	Dyke	110/78	114.47.3	39.9.15	585	**1**, 0.95
83	Yixian	Dyke	245/807	115.25.0	39.28.25	585	**2**, 1.94
84	Yixian	Dyke	232/86	115.25.47	39.26.42	580	**1**, 0.97
85	Yixian	Dyke	207/847	115.25.33	39.26.57	–	**2**, 1.45
86	Yixian	Dyke	210/78	115.21.5	39.26.15	570	**2**, 1.80
87	Yixian	Dyke	55/85	115.21.11	39.26.6	585	**1**, 0.91
88	Yixian	Dyke	25/40	115.15.47	39.23.49	–	**2**, 1.30
89	Yixian	Dyke	330/75	115.17.9	39.21.48	585	**1**, 0.90
90	Yixian	Dyke	290/74	115.15.53	39.20.15	580	**1**, 0.91
91	Yixian	Dyke	302/83	15.15.53	39.20.15	585	**1**, 0.89

M = metamorphic country rock. Type **1** and **2** thermomagnetic curves are defined in the text and RM is ratio of M_s at 100 °C on cooling to the value at this temperature during heating.

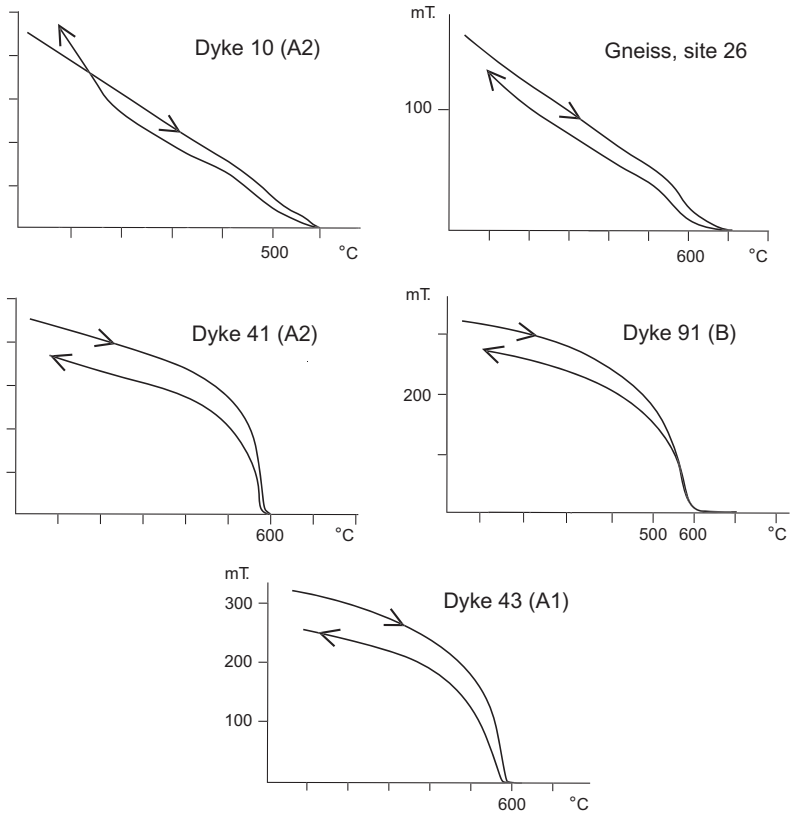


Fig. 4. Examples of thermomagnetic (Saturation magnetisation, M_s, versus temperature) spectra in dykes and metamorphic rocks of this study. The first example is Type 2 behaviour and the remainder show Type 1 behaviours with clear inflections near the Curie point of pure magnetite.

heating whilst others show large RM values. Sites exhibiting these properties are rare in the north but become dominant towards the peripheries of the study area and especially towards the SE into the region of subsequent block faulting (Fig. 3). An implication of thermomagnetic analysis is that Type 2 sites are the signature of later alteration that is most important in the SE where it has partially or completely broken down primary titanomagnetites into ferromagnesian silicates and hematite.

Signatures of the dykes to laboratory remanence acquisition comprised single lines of positive slope, the signature of paramagnetism, variably modified by a hysteresis loop, the signature of ferromagnetism, either with pot-bellied shapes indicative of one

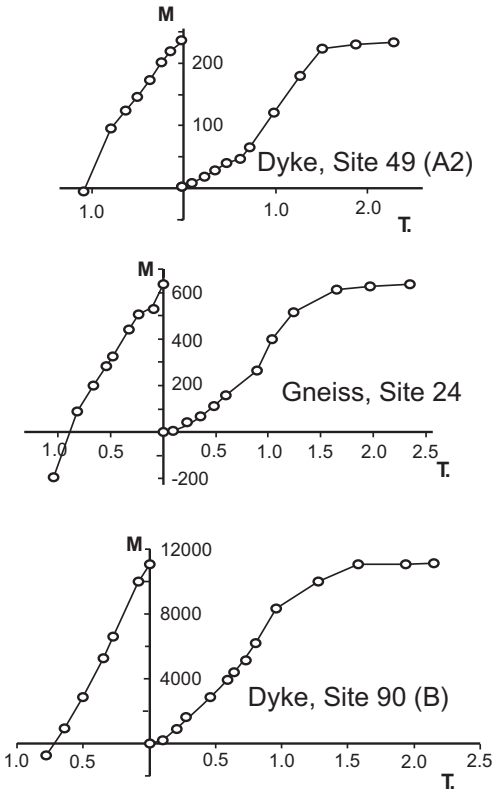


Fig. 5. Examples of IRM spectra from samples of this study.

dominant ferromagnetic component or wasp-waisted indicating superimposed effects of more than one ferromagnetic component. Loop shape is a function of ferromagnet mineralogy and domain state: titanomagnetites and titanomaghemites saturate in fields of 300 mT or less, whilst titanohematites require much larger fields to achieve saturation. Parameters used to express these properties are saturation magnetisation, M_s, saturation remanence, M_{rs}, and coercive force, H_c with $X_{MD} = (0.5 - (M_{rs}/M_s))/0.48$ yielding a semi-quantitative estimate of MD grain fraction when magnetite is dominant (Thomas 1993). Both M_{rs}/M_s and X_{MD} values indicate predominant MD sources in the dykes although coercivities tended to be higher than the typical value of $\sim 1.7 \times 10^3$ A/m for pure MD magnetite (O'Reilly 1984) and record the minority influence of SD or PSD carriers presumable responsible for the stable magnetic remanence. The uniform feature of the VFTB results is for samples with Type 2 thermomagnetic behaviour to show prominent paramagnetism and suppressed hysteresis whereas the latter is dominant in Type 1.

IRM study to fields up to 3.1 Tesla progressively imparted by a pulse magnetizer is able to resolve more clearly the presence of high coercivity ferromagnets such as hemo-ilmenite and three typical examples are illustrated in Fig. 5. Although saturation is largely achieved by 1–2 T, the IRM increments irregularly and the absence of a pure low coercivity titanomagnetite spectrum below ~300 mT. suggests that degrees of incipient alteration of this mineral to hematite are ubiquitous in these dykes. The distributed coercivity spectra are evidently a reflection of the coronal alteration in these dykes (Peng et al., 2007). Although primary igneous minerals including ortho- and clino-pyroxenes and (locally) olivine have survived, they illustrate several stages of retrogression including those analogous to reactions under granulite-facies during late

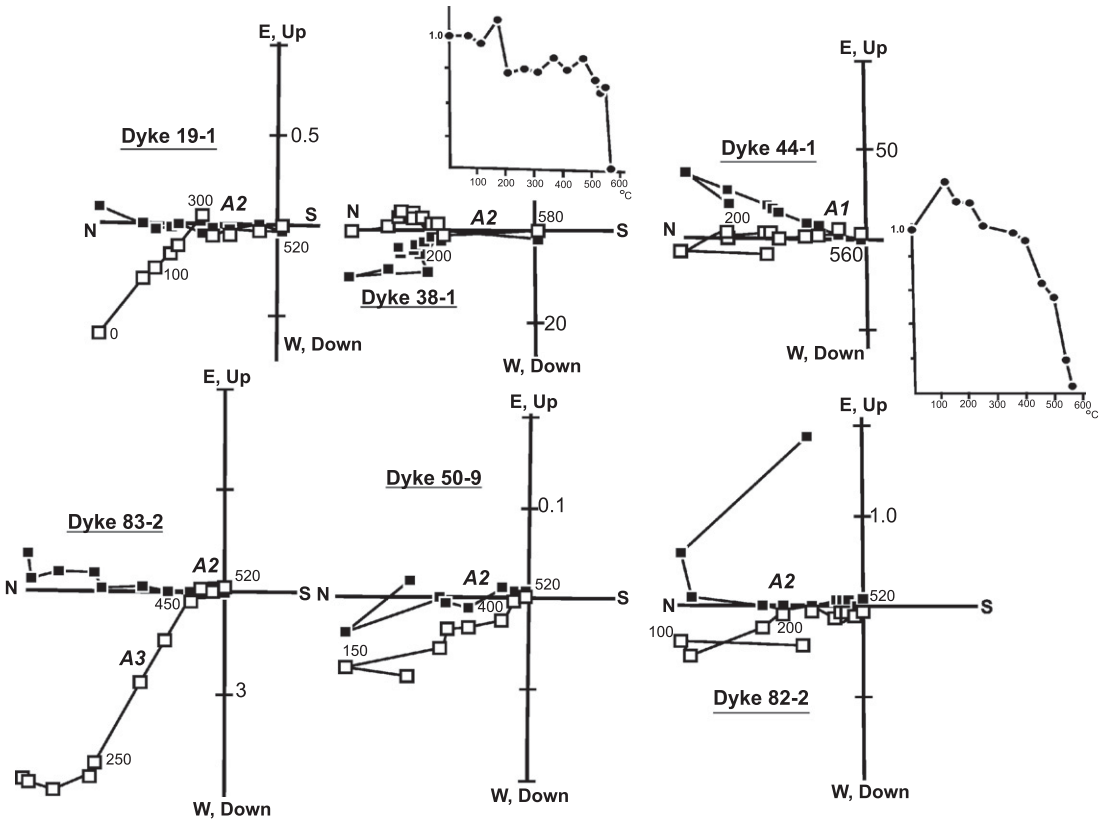


Fig. 6. Examples of dyke palaeomagnetic samples with A1 and A2 components of demagnetisation; examples 19–1 and 83–2 have distributed A3 components removed to recover A2 components. Closed squares are projections of magnetisation vectors onto the horizontal plane and open squares are projections onto the vertical plane. Figures on the axes are magnetisations in 10^{-5} A m^2/kg.

phases of magmatism (Markl et al., 1998). Peng et al. (2007) identify three stages of metamorphism in the dykes which they attribute to alteration during uplift from depths of up to 20 km and which are evidently responsible for the broadened hysteresis spectra.

5. Palaeomagnetic results

Sixteen sites exhibited either unstable behaviors or showed no coherent within-site groupings during thermal demagnetization and are not considered further. The remainder have demagnetization trajectories typically showing two or three component behaviors, sometimes following removal of a viscous remanence in the first step or two of treatment, and with the lower unblocking temperature component having steeper inclination than the higher one. Characteristic Remanent Magnetisations (ChRMs) most commonly have N to NNE declinations. A population with shallow (<20°) inclinations is distributed around the primitive between N and NNE. By analogy with previous palaeomagnetic studies of these dykes (Qian and Chen, 1987; Halls et al., 2000; Hou et al., 2001) we refer to magnetisations with declinations >25° as A1 and magnetizations with declinations <25° as A2. Demagnetisation trajectories for these components are typically distributed and convergent (19–1, 44–1, 50–9, 82–2 in Fig. 6, 75 in Fig. 9) with rare examples of opposite polarity, two belonging to A1 and two to A2 (dyke 61 in Fig. 7).

A1 magnetisations are comparable to the dominant population of directions observed in dykes emplaced into the granulite belt to the north of this study area investigated by Qian and Chen (1987) and Halls et al. (2000) and also observed in a single dyke from the same terrane by Zhang and Piper (1994). Only 4 dykes in the present collection from the region to the south of these earlier studies show A1 directions, although the two reversed members found here show that cooling and magnetisation of this group actually occurred during an interval including polarity reversal. These four dykes all have Type 1 thermomagnetic curves indicating minimal alteration and a primary cooling-related magnetisation. The larger population of shallow inclination magnetisations in this study has northerly shallow directions of ChRM belonging to A2 and demagnetisation behaviours are mostly comparable to A1; they are typically the higher unblocking temperature component of multi-component structures recovered below the Curie point of magnetite (Figs. 6–8). This magnetisation comprised 3 of 4 dykes studied by Qian and Chen (1987) and was a minority group (7 sites) in the population studied by Halls et al. (2000). Like A1, the A2 group is dominated by one polarity although the interval of magnetisation is now recognised to have included polarity reversal. A2 dykes tend to have distributed spectra although some have narrow unblocking temperature spectra (82-2, 83-2 in Fig. 6 and 61-7 in Fig. 7 and 64 in Fig. 10) and the majority (70%) exhibit Type 1 thermomagnetic curves. The predominant strike trend of these A2 dykes is northerly (10°E) although a third have westerly trends in common with 3 of the 4 A1 dykes identified here; it appears that both northerly and westerly trends are conjugate members of the same dyke system as indicated by age dating (Peng et al., 2005, 2007).

The A1–A2 group merges into dykes with comparable declinations but steeper positive inclinations ($I > 20°$) and referred to as A3 (Figs. 7, 9 and 10). These ChRMs mostly have N-NNE declinations and intermediate positive inclinations close to, but shallower than, the direction of the present geomagnetic field in the sampling region (Fig. 11); two examples are of opposite polarity. The similarity of these directions to the Mesozoic and later field and more variable demagnetisation behaviours renders this population difficult to categorise. Some show discrete and convergent unblocking temperature spectra (dykes 10 and 72 in Fig. 9) whilst others fail to converge (dyke 54 in Fig. 10) and/or may unblock to recover A2 in the highest part of the unblocking temperature spectrum

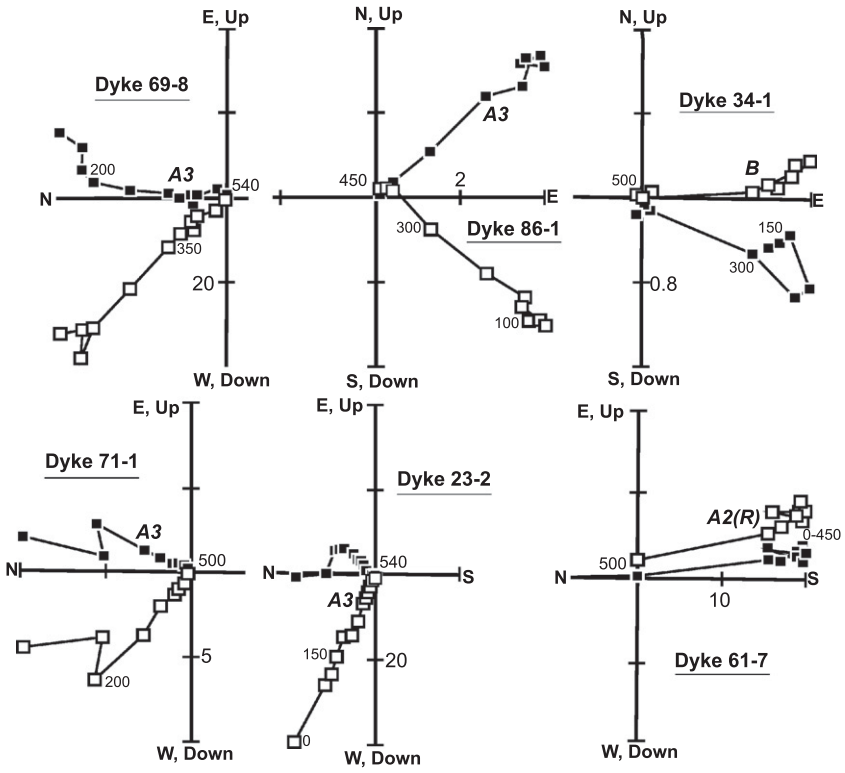


Fig. 7. Examples of palaeomagnetic results recovering a reversed A2 and a B magnetisation (34-1 and 61-7) with four others dominated by distributed A3 components. Symbols are as for Fig. 6.

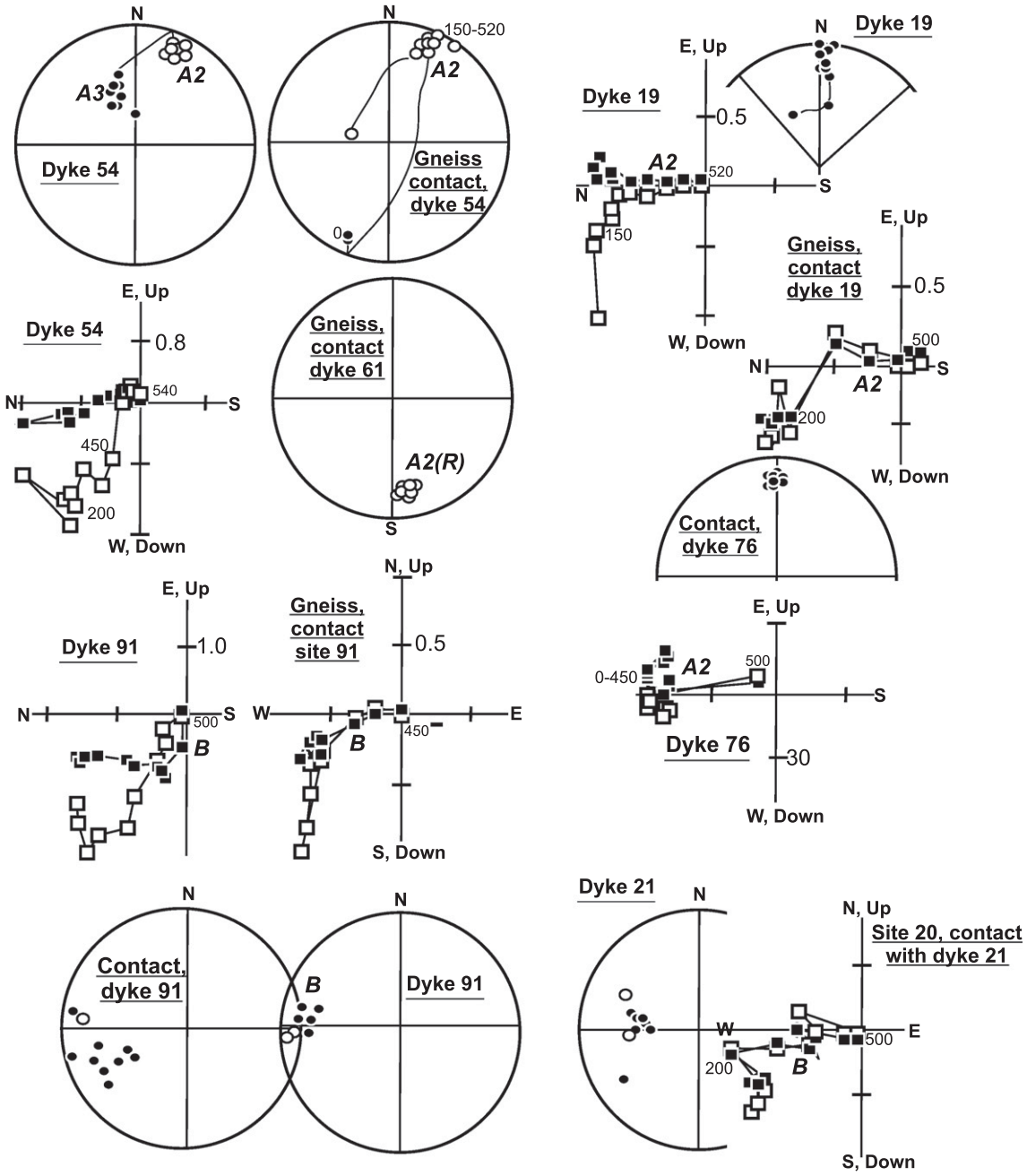


Fig. 8. Examples of demagnetisation results recovering A2 and B magnetisations at dyke contacts with gneissic country rocks. Symbols are as for Fig. 6.

(83-2 in Fig. 6, 69-8 in Fig. 7 and dyke 64 in Fig. 10. The single magnetite-resident components show convex intensity decrease during demagnetisation whilst other examples show a large low unblocking temperature fraction with little or no convex step near to the Curie point (c.f. dykes 10 and 72 in Fig. 9). It is likely that these are partial or complete overprints of A1 and A2, a point confirmed in some dykes where these latter magnetisations are recovered in isolated cores in the higher part of the demagnetisation projections. Supporting the view that overprinting has occurred as a result of much later Mesozoic-Tertiary alteration, 50% of the A3 sites show Type 2 thermomagnetic curves. In addition these dykes have comparable strike trends to A1–A2 (c.f. Tables 1 and 2) with the majority being northerly (averaging 25°E) and about a third having easterly trends (280°E). Nevertheless Type 1 curves and convergent magnetisations may also be expected if the dykes are primary and Mesozoic or Tertiary in age, and Dyke 68 is an example of a reversed equivalent magnetisation close to the probable thermal influence of a Mesozoic granite. In contrast sites between 68 and 81 in the Fuping Inlier trending between NW and NE have Type 1 spectra and directions that merge with A2 directions from dykes of similar trend in the same region: these could either be contiguous with the A1–A2 population or much later magnetisations.

The mean direction derived from A3 dykes showing Type 1 thermomagnetic spectra is only marginally different (D/I = 12/46°, 15 dykes) from dykes exhibiting Type 2 spectra (D/I = 14/41°, 14 dykes). Hence we interpret A3 as a composite population: some could be a temporal continuation of the A1–A2 group but these are inseparable from Mesozoic-Recent primary and overprinted components. Directions comparable to A3 also dominate the country

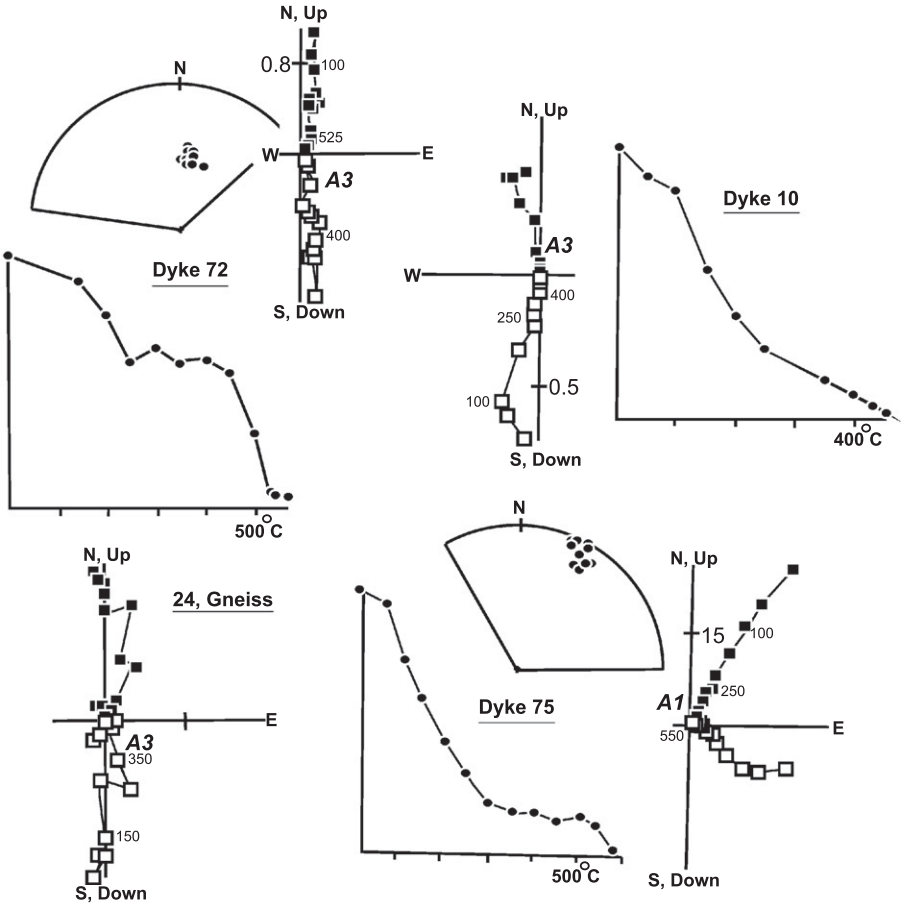


Fig. 9. Examples of orthogonal projections and demagnetisation spectra from dykes and country rock exhibiting A1 and A3 components. Symbols are as for Fig. 6.

metamorphic rocks in this region (contact samples in Fig. 9 and sites 26 and 53 in Fig. 10, Table 2); this is in contrast to the granulite terranes to the north where magnetisation similar to A1 and A2 characterise the metamorphic basement (Zhang and Piper, 1994). Small numbers of type A3 magnetisations were also found by Halls et al. (2000) further to the north and considered to be either tilted versions of A1 or Phanerozoic overprints. Since the sampled A1 and A2 dykes are vertical to subvertical intrusions into metamorphic basement rocks, we have recognised no evidence to suggest that they not post-tectonic; we are unable to exclude small amounts of rotation during Mesozoic and later faulting although the tight groupings of ChRM directions achieved by the component analysis (Fig. 12) suggests that such effects are likely to be small.

Finally a dual polarity population of 11 dykes has ChRM directions with contrasting easterly and westerly declinations (Fig. 11) referred to as 'B' and directed westerly (6 dykes) or easterly (5 dykes) in the highest unblocking temperature segment of two and three component behaviors resident in magnetite (34–1 in Fig. 7). The majority (70%) have Type 1 thermomagnetic spectra whilst samples in contact gneiss at site 91 and at the contact with dyke 20 identify this magnetisation as a probable thermal overprint related to the dyke emplacement (dykes 21 and 91 in Fig. 8). The strike trends of these dykes are mostly northerly like the A1-A2 population averaging 5°E with 3 dykes having westerly trends (Table 1). Single examples of this dyke population were found in previous studies by Zhang and Piper (1994) and Halls et al. (2000) with the latter example in baked host rocks. The mean of sites 4, 5, 6, 20, 29, 30, 32, 34, 67, 90 and 91 with this ChRM axis is D/I = 108/7°; this is quite different from both the other dyke populations of this study and from the Mesozoic-Recent field direction in North China.

Contact tests to help establish the age of the magnetizations had variable success due to limited magnetic stability in the metamorphic country rock and MD properties of some dykes selected for testing, but they successfully link A1 and A2 to probable primary igneous cooling (Fig. 8). Gneiss samples at the contact of dyke 19 for example remove a steeper magnetization analogous to A3 to recover A2 whilst samples away from the dyke are characterized by A3 indicating that this is pervasive in facies more susceptible to remagnetisation (dyke 54 in Figs. 8 and 10); similar magnetizations are additionally found in nearby gneisses at sites 24 and 26. At site 54 where magnetizations similar to A3 are unblocked to recover A2, samples from the contact gneiss record A2 (Fig. 8); sample 61-1 in this figure is a gneiss supporting a primary cooling-related origin for a reversed polarity example of A2, whilst the contact with site 76 has a narrow unblocking temperature recording overprinting by normal polarity A2. The contacts with A3 are unfortunately, ambiguous because this is also the dominant magnetisation in the country rocks (Figs. 9 and 10). A1 is rare in this collection but positive contact tests are reported in the granulite facies terrane to the north by Halls et al. (2000).

However, as intimated by these latter authors, these tests may also be ambiguous because magnetisations in the basement here are similar to A1 and A2 (Fig. 13). Nevertheless their study satisfactorily constrained the A1 axis, reinforced by the reversal recognised in this study, to their single-grain zircon U–Pb isochron age of 1769.1 ± 2.5 Ma presumed to be the age of emplacement of

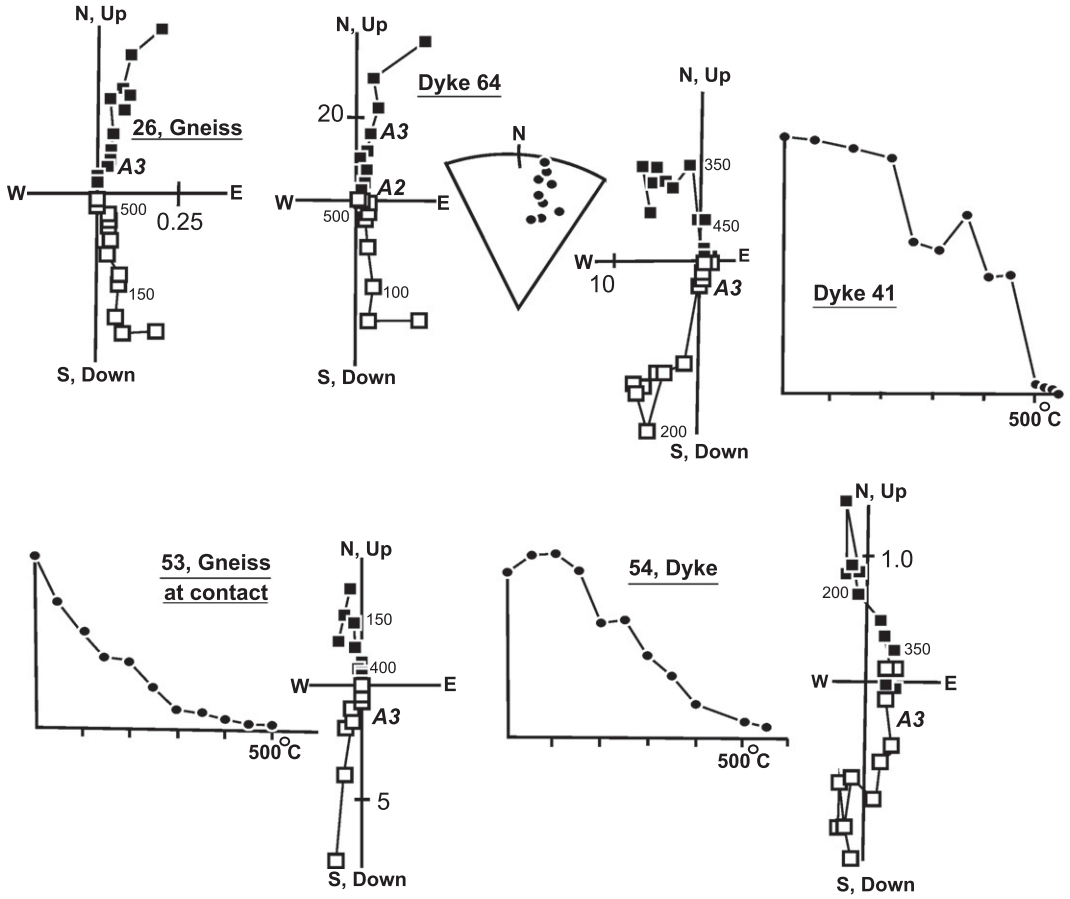


Fig. 10. Comparative demagnetisation spectra and orthogonal projections in A3 components from dykes and gneiss contacts; although spectra tend to be distributed and dominant in this population A2 is frequently recovered in the highest part of the magnetite unblocking temperature spectra (e.g. dykes 64). Symbols are as for Fig. 6.

the A1 dykes. Since the A2 population is contiguous we infer that it has a similar age to A1; the strike trends of A2 range from NW to NNE with some conjugate members (Fig. 11). No detailed comments on palaeomagnetic results from the metamorphic country rocks are merited here: with one exception (site 14) they are of northerly positive direction and comparable to magnetizations observed in the East Miyun, Qinglong and North Zhunhua terranes of the North China Shield (Piper and Zhang, 1999). Their proximity or coincidence with the A3 group of magnetizations tends to support the view that they record a regional remagnetisation event linked either to late stages of ~1780–1760 Ma dyke emplacement, and/or much later Mesozoic-Tertiary magmatic and tectonic activity.

6. Interpretation

The palaeomagnetism of dyke swarms in the NCS has formerly been studied by Qian and Chen (1987), Halls et al. (2000) and Hou et al. (2001) with a few examples included in a study of the granulite basement by Zhang and Piper (1994); dykes in the Luilang Inlier to the west were also included in an investigation by Hou et al. (2001) who report some data in common with Halls et al. (2000). The present study greatly expands the number of dykes investigated and extends investigation mainly to the south where they are emplaced into country rocks of lower metamorphic grade. Results from the collective studies are summarized as group mean directions and pole positions in Table 3; site mean directions are plotted in Fig. 12 and the regional distribution of magnetization type is shown in Fig. 14.

Northerly shallow directions comparable to A1 and A2 are found in the high grade (granulitic facies) metamorphic terranes to the north and west of the present study area (Section 2, Zhang and Piper 1994). Although the metamorphic magnetizations could be attributed to regional remagnetisation by the A1–A2 dyke emplacement, we consider this unlikely because the remanence is found remote from the dykes in high grade basement where it is fixed in high unblocking temperature magnetizations within metamorphic magnetite blades and needles showing SD or PSD properties (Zhang and Piper 1994). The granulite magnetisations comprise a long dual polarity swathe (plotted with common polarity in Fig. 13) related to late stage uplift-related exhumation and cooling. This is the regional magnetization in the most deeply-exposed sector of the NCS in this region and is also where the feldspars show the highest degrees of clouding (Halls et al., 2000).

The precipitation of submicroscopic magnetite in feldspars is generally considered to be a response to slow cooling at deeper crustal levels and such magnetite grains are typically the carriers of high unblocking temperature stable remanence residing in single domains (Morgan and Smith, 1984; Halls and Zhang, 1998). Remanence in the granulite facies terrane has been shown by Zhang and Piper (1994) to be resident in multiple phases of magnetite precipitation in micro-fractures during uplift decompression postdating deformation and metamorphism. In the Inner-Mongolia-North Hebei Belt it is constrained by U–Pb age evidence in the range 2.0–1.8 Ga on granulite to amphibolite facies basement gneisses (Zhai et al., 1996, Li et al., 1997; Guo and Shi, 1996). Hence it is probable that exhumation of the granulite basement and the A1–A2 dyke emplacement, uplift and cooling have recorded the

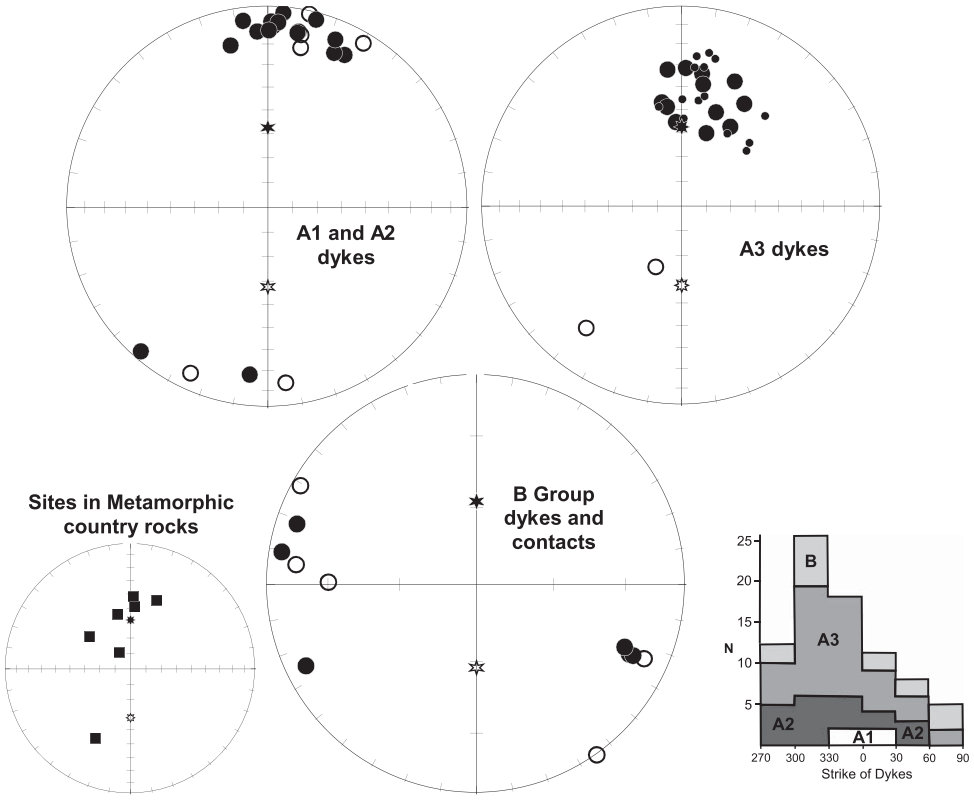


Fig. 11. Site mean directions of magnetisation from dykes and metamorphic rocks of the North China Shield; solid symbols are lower hemisphere plots and open symbols are upper hemisphere plots. The A3 magnetisations derived from sites showing Type A thermomagnetic spectra are plotted as larger circles and A3 magnetisations derived from sites exhibiting Type B spectra are plotted as smaller circles. The star is the direction of the average geocentric dipole field in this region.

same segment of APW at ~1.80–1.76 Ga. The similarity of A1–A2 to magnetizations in the granulite basement (Fig. 13) would then be a reflection of the uplift decompression; the extensional regime and dyke emplacement are all inferred to have occurred relatively quickly in geological terms following the peak of granulite metamorphism at ~1.85 Ga.

The small number of dykes identified here carrying the A1 remanence are in the vicinity of the city of Fuping where they trend between NNE and NNW (340–10°E) and lie on the SSE extension of the NNW trending dykes carrying this same remanence in the Datong region (Figs. 2 and 14). All dykes so far identified with this remanence appear to have northerly to north westerly strike. Dykes with the northerly declinations included in the A2 group occur in the same region, as well as on either side, of this A1 axis. The migration in declination embraced by the swathe between them (Figs. 12 and 13) is presumably therefore a reflection of APW rather than a signature of tectonic rotation. The A2 population of palaeomagnetic directions also includes some dykes with W-WSW strike; Peng et al. (2008) note examples of comparable U–Pb age determinations between dykes trending N–S and E–W and these are therefore likely to be conjugate members of the same emplacement event.

An increase in feldspar clouding in the dykes and decrease in hydrous alteration towards the north imply that the NCS has been tilted to the south (Halls et al., 2000) with this latter study showing three positive baked contact tests with dykes having the A1 direction whilst the unbaked host rock showed A2, thus implying that A2 is older. The dominance of the A2 component in dykes towards the south recognised in this study (Fig. 14) also suggests that this remanence is older since the sampling sites in these dyke would have been at higher levels in cooler crust. Since the A1–A2 remanence characterises the crust as a whole (Zhang and Piper 1994), we interpret this remanence in the dykes as of thermo-chemical origin and acquired during uplift-related cooling (Peng et al., 2007 and Section 4); this would accommodate the similarity between the dyke and metamorphic basement magnetisation recorded in the north (Fig 13). The dykes characterised by A1 magnetisations presumably remained hotter and cooled more slowly in deeper crust compared to the shallower A2 dykes. As a result they would have acquired their remanence at variable times later than A2 as the NCS remained in low latitudes and rotated clockwise (Fig. 12). The A1 dykes typically have the most stable remanence, a characteristic acquired by the growth of micron-sized exsolved magnetite in the feldspars (Halls et al., 2000); this would also produce the signature dominated by Type 1 thermomagnetic curves, higher coercivity and dominant ferromagnetic hysteresis signature (Section 4).

The A3 dykes are the most widely distributed and are concentrated within the amphibolite facies terranes in the southern part of the NCS. Thermomagnetic signatures of ferromagnet alteration (Type 2 curves) are more prominent and stabilities to demagnetisation are more variable. Whilst the composite nature of this group does not merit more detailed interpretation, we observe that Type 2 signatures of alteration are most prominent near the peripheries of the Proterozoic inliers (Fig. 3) and are therefore possibly linked to alteration associated with Mesozoic and later faulting in these areas; supporting this interpretation, A3 poles lie fairly close to Late Jurassic-Cretaceous poles from the NCB (Gilder and Courtillot 1997; Uno and Huang 2003). The B dykes however, are evidently an integral swarm of Precambrian age because their direction of magnetization corresponds to no known Phanerozoic palaeofield for North China. From a range of geologic, petrologic and isotopic evidence Shao et al. (2005) recognize three distinct episodes of dyke emplacement into this central sector of the NCS in addition

Table 2
Site mean Palaeomagnetic results, dyke swarms and basement rocks of the North China Shield.

Site no.	Rock type	*D*	*I*	*N*	*R*	α_{95}	*k*
2	Dyke[3]	49.5	55.0	4	3.87	19.7	22.7
3	Dyke[3]	9.0	45.6	7	6.70	13.7	20.2
4	Volcanics[B]	114.5	21.3	10	9.77	7.9	38.6
5	Dyke[B]	114.4	19.3	11	10.51	10.3	20.5
6	Dyke[B]	112.9	24.5	7	6.69	14.1	19.2
7	Dyke[2]	9.9	12.5	8	7.70	11.7	23.4
8	Dyke[2]	10.1	23.3	10	9.76	8.0	37.5
9	Dyke[2]	1.6	11.0	6	5.84	12.0	31.9
10	Dyke	0.3	45.6	7	6.84	10.0	37.6
12	Gneiss[3]	69.0	39.4	12	11.70	7.3	36.4
13	Amphibolite[3]	307.5	56.3	6	5.95	6.8	98.3
14	Amphibolite[3]	206.7	37.9	5	4.82	16.5	22.4
15	Gneiss[3]	324.9	77.3	8	7.85	8.2	46.6
16	Dyke[3]	218.1	−23.7	3	2.98	10.9	128.3
17	Dyke[3]	328.8	43.7	4	3.90	17.1	29.8
18	Gneiss	1.6	42.1	7	6.61	16.0	15.3
19	Dyke[2]	1.0	8.5	4	3.98	7.6	147.3
20	Dyke[B]	255.5	11.0	7	6.66	14.9	17.4
23	Dyke[3]	18.6	58.4	7	6.88	8.5	50.9
24	Gneiss[3]	21.3	41.1	5	4.89	12.7	37.0
26	Gneiss[3]	3.8	49.3	9	8.96	3.7	191.8
27	Dyke[3]	5.7	25.7	4	3.96	10.4	79.6
28	Dyke[2]	356.5	13.6	5	4.88	13.2	34.6
29	Dyke[B]	299.3	−4.4	9	89.0	5.7	84.0
30	Dyke[B]	279.5	6.7	3	3.00	5.3	544.7
31	Dyke[2]	23.2	17.7	5	4.90	12.3	39.5
32	Dyke[B]	144.8	−0.9	5	4.93	10.3	56.6
33	Dyke[3]	320.5	37.0	8	7.79	9.6	33.9
34	Dyke[B]	113.9	−13.6	12	11.86	4.8	81.4
35	Dyke[3]	32.1	54.8	5	4.94	9.1	70.9
36	Dyke[3]	46.9	51.8	4	3.97	10.0	85.8
38	Dyke[2]	4.5	3.5	5	4.91	11.6	44.3
40	Dyke[3]	8.7	33.5	5	4.81	17.0	21.2
41	Dyke[3]	349.1	46.2	7	6.86	9.3	43.5
42	Dyke[2]	205.0	−10.0	7	6.75	21.7	23.7
43	Dyke[1]	30.2	−6.4	7	6.98	3.6	283.3
44	Dyke[1]	26.7	16.4	8	7.90	6.5	73.5
45	Dyke[2]	9.6	13.2	14	13.82	4.7	71.3
46	Dyke[1]	221.4	5.1	6	5.96	5.9	131.0
47	Dyke[3]	184.5	−64.7	8	7.95	4.5	150.8
49	Dyke[2]	12.1	−2.0	6	5.60	19.8	12.3
50	Dyke[2]	347.0	18.6	6	5.84	12.1	31.7
51	Dyke[3]	351.5	48.5	6	5.93	8.3	66.3
52	Dyke[3]	5.3	31.0	8	7.79	9.7	33.8
53	Gneiss[3]	346.6	52.6	8	7.74	10.9	26.7
54	Dyke	10.9	−13.9	4	3.94	13.4	14.8
55	Dyke[3]	9.1	30.3	6	5.72	16.4	17.6
56	Dyke[2]	11.6	−20.2	5	4.82	16.7	22.1
58	Dyke[B]	254.1	−42.0	7	6.94	6.1	97.7
61	Dyke[2]	174.2	−13.4	8	7.88	7.2	60.0
62	Dyke[2]	186.2	17.5	4	3.91	15.9	34.2
63	Dyke[3]	12.8	25.5	8	7.76	10.3	29.7
64	Dyke[2]	3.2	8.8	5	4.87	13.8	31.7
67	Dyke[B]	288.6	10.8	6	5.54	21.2	10.9
68	Dyke[3]	203.2	−63.0	5	4.97	6.9	124.3
69	Dyke[3]	9.9	38.2	7	6.85	9.7	39.3
70	Dyke[3]	1.7	31.5	4	3.96	11.3	67.0
71	Dyke[3]	31.4	51.6	5	4.95	8.9	75.5
72	Dyke[3]	20.0	48.6	6	5.92	8.4	64.7
74	Dyke[3]	356.0	55.4	6	5.82	12.9	28.0
75	Dyke[2]	21.9	11.2	7	6.77	12.1	25.7
76	Dyke[2]	0.3	13.1	8	7.89	6.9	65.5
77	Dyke[2]	14.4	4.3	7	6.83	10.4	34.4
78	Dyke[3]	31.4	39.9	7	6.95	5.7	115.1
79	Dyke[3]	23.0	32.8	7	6.97	4.2	204.3
80	Dyke[3]	347.0	47.6	7	6.92	7.2	71.0
81	Dyke[3]	354.0	32.0	7	6.81	11.0	31.2
82	Dyke[2]	352.4	7.3	6	5.81	13.5	25.7
83	Dyke[3]	1.3	53.9	6	5.89	9.8	47.3
84	Dyke[B]	81.1	−36.1	7	6.82	10.6	33.2
85	Dyke[3]	11.7	43.3	8	7.79	9.7	33.3
86	Dyke	42.7	38.6	5	4.93	10.3	56.0
87	Dyke	86.5	81.0	8	7.95	4.5	150.2
88	Dyke	16.3	82.4	6	5.90	9.5	50.5
89	Dyke	282.1	−82.6	2	2.00	9.6	672.3
90	Dyke[B]	270.9	−30.2	6	5.89	9.9	46.9
91	Dyke[B]	276.3	−14.7	7	6.57	16.7	14.0

D and *I* are the mean declination and inclination of the magnetisation derived from N samples yielding a resultant vector of magnitude *R* and Fisherian precision, *k* (=$(N-1)/(N-R)$). The superscripted assignments in the second column are the magnetisation assignments used in this study.

to the most populous ~1.78–1.76 Ga episode: dykes dated ~0.8–0.7 Ga have variable, mostly NE trends. A prominent Mesozoic episode dated ~153–135 Ma includes dykes trending NW to NNE, and a few Early Cenozoic dykes (~50 Ma) also have NW trends. The two Phanerozoic palaeofields are compatible with A3 magnetisation directions but are incompatible with the B dyke magnetizations. This reinforces the view that the B dykes are Precambrian in age and although they differ from known 950–650 Ma magnetizations from the NCS (Zhang, 1998), Meso-Neoproterozoic APW is poorly constrained by magnetizations of firmly-established primary origin, and these dykes could well belong to this era; they are evidently appreciably younger than ~1.8–1.7 Ga, the oldest age of magnetization recorded in the NCS.

7. The ~1.78 Ga LIP event and continental crust in Palaeoproterozoic times

The wider significance of the APW swathe from North China is evaluated here in terms of the quasi-rigid supercontinent Palaeopangaea (Piper, 1982, 2007). The primary case for this is recent evaluation in the context of the mid-Archaean-Palaeoproterozoic (~2.9–2.0 Ga) palaeomagnetic database (Piper, 2010a) where the quasi-static character of APW during the 500 Myr interval 2.7–2.2 Ga Archaean-Early Palaeoproterozoic identifies crustal assemblages recognized by Rogers (1996) and Rogers and Santosh (2004) constrained to a symmetrical hemispherical continental reconstruction analogous to Wegener's (Neo)Pangaea. The long-term persistence of this reconstruction through Proterozoic times is demonstrated by continuing conformity of 1.3–0.6 Ga palaeomagnetic poles to a single APW path comprising successively the Gardar Track, Keweenawan Track, Grenville-Sveconorwegian Loop and Franklin-Adelaide Track and culminating in demise of the supercontinent at ~0.6 Ga (Piper, 2007, 2010b). The latter break-up event is identified by diverse geological signatures near the dawn of the Cambrian (Bond et al., 1984; Condie, 1997; Rogers and Santosh, 2004). Key facets of the Palaeopangaean reconstruction are eminently testable between 2.7 and 2.2 Ga (represented by a substantial number of radiometrically-constrained poles from igneous units) and after 1.3 Ga (from a more extensive database) and provide justification for testing Palaeopangaea in place of the popular 'Columbia' model. An evaluation during the late Palaeoproterozoic has been restricted mainly to the palaeomagnetic record from Hudsonian and Svecofennian tectono-thermal terranes of Laurentia and Fennoscandia where results are predominantly of post-orogenic uplift-related origin; APW has been correspondingly difficult to define and date although some key APW trends have been noted (Morgan, 1976, Piper, 1982, 1985; Mitchell et al. 2010).

The recognition of a major LIP event at ~1.8 Ga enables the validity of the reconstruction to be addressed with more confidence because investigations during the past two decades have produced a range of poles from igneous rocks, including a number linked to precise age determinations. The ~1.85–1.75 Ga data are

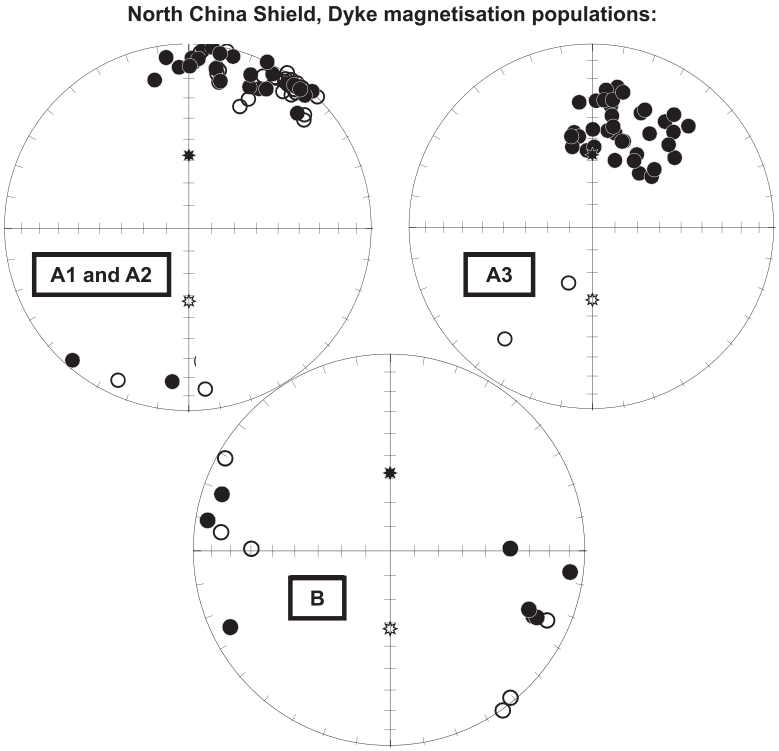


Fig. 12. Summary of populations of dyke site mean directions belonging to the A1, A2 and B directions from palaeomagnetic studies in the North China Shield. Mean directions and pole positions are listed in Table 3.

summarized in Table 4 and when rotated to Palaeopangaea parameters resolved from ~2.7 to 2.2 Ga data (Piper, 2010a,b) conform to a swathe crossing the supercontinent (Fig. 15). The analysis is reinforced by the dominant common polarity (closed fraction of the symbols) which is also evident in the dykes from the NCS and supports the view that poles of the same polarity from diverse shields are being correctly correlated with one another. The distribution of assigned ages does not allow the APW direction to be resolved with certainty although age assignments to poles north of 35° and east of 250° in Fig. 15 average ~1.77 Ga compared with an average age of 1.82 Ma for poles south and west of this region suggesting that APW is S → N conforming to the age sequence A2 → A1 resolved from the NCS. In addition, the linked small stars in Fig. 15 show the polar sequence derived from the NCS granulite terrane (c.f. Fig. 13); these magnetizations postdate the peak of granulite metamorphism at ~1.83 Ga and seem to be temporally correlated with the A1–A2 dyke record although the sense of migration within the granulite data is not unambiguously resolved. This APW swathe from the NCS connects the southerly and northerly polar groups in Fig. 15 and reinforces the interpretation that a common APW trend is being resolved here.

The position of Guyana-Amazonia in supported by the 1789 ± 7 Ma pole from the Colider volcanics (Bispo-Santos et al., 2008) with two poles from dolerites probably linked to the ~1.8 Ga LIP event plotting further north within the younger limit of the swathe (Table 4 and Fig. 15); a spatial link between the Sao Francisco, Guiana and West Africa cratons is suggested by common geological histories (Zhao et al., 2002a,b, 2004a,b,c; Rogers 1996) but no palaeomagnetic data are yet available to test the Palaeoproterozoic configuration of the East Antarctica Shield. The Mashonaland Dolerite pole from the Zimbabwe Craton is removed to the west of the remaining data; although the age of this unit is poorly defined, it plots near the beginning of the APW swathe from the NCS (Fig. 15) in a correlation reinforced by older and younger poles from Southern Africa with the Laurentian record on this same reconstruction (Piper, 2007, 2010a,b). Although results listed in Table 4 are restricted to igneous rocks, we note that poles from Australia inferred to be primary from the ~1.8 Ga Kimberley Group (Schmidt and Williams, 2008) accord with the poles 1, 2 and 4 from Australia (Table 4) and yield correspondingly similar poles within the distribution of Fig. 15. A single pole from the Dharmapuri dykes of India falls near the northerly extension of the APW track correlating with the pole from the A1 dykes in the NCS and suggests a possible correlation with other proto-Gondwana shields. The significance of a 1.855 Ga ^{39}Ar–^{40}Ar age on these dykes is unclear: the E–W trending dykes in the Dharwar craton are substantially older at ~2.2 Ga in age (French and Heaman, 2010) although circumstantial evidence indicates dyke emplacement in South India at ~1.8 Ga prior to commencement of Cuddapah sedimentation at about this time (Bhaskara Rao et al., 1995, Radhakrishna and Joseph, 1996).

Proximity between the NCS and India during the Proterozoic is consistent with the geological record since they appear to share a number of unique events (Kröner et al., 1998; Zhao et al., 2003) the earliest being a major Archaean crust-forming event at 2.6–2.5 Ga accompanied by near-contemporaneous granitioid emplacement with metamorphic peak attained shortly after (<50 Ma) granite intrusion (Kröner et al., 1998). Dome-and-basin structures are dominant in both terranes and anticlockwise P–T–t paths characterize metamorphic evolution of the Late Archaean granulite host terranes (Zhao et al., 2002a,b, 2003). Both shields have similar Palaeoproterozoic formations comprising lower clastic-rich, middle volcanic-rich and upper clastic + carbonate sequences (Zhao et al., 2003; Mazumder 2003); post-tectonic dyke emplacement and commencement of Cuddapah sedimentation in the South India Shield also occurred at ~1.8 Ga at about the same time as the initiation of aulacogen formation in the NCS (Bhaskara Rao et al., 1995, Zachariah et al., 1999). Of several geometric links

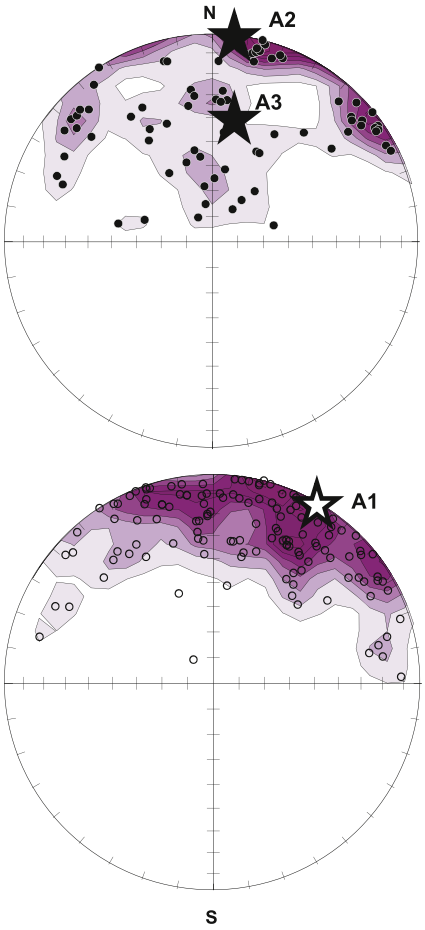


Fig. 13. Directions of magnetisation in the granulitic basement rocks of the North China Shield after Zhang and Piper (1994) compared with mean directions derived from A1, A2 and A3 dyke populations.

proposed between the NCS and other cratons during the Palaeoproterozoic (Condie 2002, Wilde et al., 1997, Zhao et al., 2003), the model of the latter authors is most closely supported by the palaeomagnetic evidence although the position of the NCS is inverted relative to the Indian Shield as required by the positions of their palaeopoles in Fig. 15; the proposed geological correlations remain essentially valid. The NW–SE trending Dharmapuri dyke swarm in the South India craton (1700–1800 Ma, K–Ar, 1855 ± 7 Ma Rb–Sr, Radhakrishna and Joseph, 1996; Radhakrishna et al., 1999) is then aligned in continuity with the NNW swarm in the North China Shield and identified as a likely component of the same ~1.8 Ga LIP event (Peng et al., 2005). Peng et al. (2007) develop a petrogenetic model for the ~1.78 Ga dyke swarm in the context of a plume-related source located at a continental margin sited near the present southern edge of the NCS. This is consistent with the palaeogeographic reconstruction of Fig. 16 and indicates that this dyke swarm, which extends up to 1000 km across the NCS, formerly extended a further ~2000 km across the Indian Shield: this suggests a dimension comparable to other Proterozoic margin-sited LIP events such as the Mackenzie (Ernst and Buchan, 1997, 2001).

The recognition of a common Palaeoproterozoic APW swathe from renewed analysis of ~1.85–1.75 Ga poles using the ~1.3–0.6 Ga core configuration (Fig. 15) supports long-term integrity of Proterozoic crust as implied by multiple signatures of continentality (e.g. Engel et al., 1974). It indicates that continental crust grew by progressive consolidation around a core of the ancient nuclei comprising 'Ur', 'Arctica' and 'Atlantica' protocontinents (Rogers 1996; Rogers and Santosh 2004) already in place by Late Archaean times (Piper, 2003, 2010a; Mondal et al., 2009). This is illustrated in Fig. 16 summarizing the orogenic belts that consolidated in Late Palaeoproterozoic times which highlights their dominant axial alignment. The NCS was evidently sited near to a periphery of the continental crust where the Trans-North China Zone includes juvenile material and was the site of eastward-thrusting over the western margin of the EB (Sun et al., 1992; Zhao et al., 2003). Central India and the NCS are the two Archaean cratons that experienced major crust-forming events between 2.6 and 2.5 Ga ago leading Kröner et al. (1998) to postulate that they constituted a single active plate margin (Fig. 16) as juvenile crust accreted onto an older nucleus. Continuity with the Capricorn Orogen of West Australia is also suggested by Fig. 16 with the latter zone formed

Table 3
Collective group mean directions and pole positions from Precambrian dyke swarms and dyke contacts in the North China Shield.

Group	Sites	*D*	*I*	*R*	α_{95}	*K*
(i) Group mean results, this study						
A1	4	30.9	3.7	3.92	15.4	36.5
		Pole position: 247.4°E, 44.1°N (dp/dm = 7.7/15.5°)				
A2	18	5.6	5.9	17.42	6.5	29.1
		Pole position: 283.4°E, 54.6°N (dp/dm = 3.3/6.5°)				
A3	28	13.7	43.2	26.83	5.3	23.1
		Pole position: 246.9°E, 72.7°N (dp/dm = 4.5/7.2°)				
B	11	108.2	6.7	10.15	13.9	11.8
		Pole position: 188.9°E, 12.1°S (dp/dm = 7.0/14.0°)				
(ii) Collective group Mean results from all dyke studies, North China Shield						
A1[1]	33	35.6	−1.3	32.54	3.0	69.6
		Pole position: 244.6°E, 38.7°N (dp/dm = 1.5/3.0°)				
A2[2]	24	7.5	1.9	22.96	6.4	22.0
		Pole position: 281.0°E, 51.3°N (dp/dm = 3.2/6.5°)				
A3[3]	38	18.2	42.7	36.22	5.2	20.8
Pole position: 240.5°E, 69.0°N (dp/dm = 4.0/6.5°)						
B[4]	14	108.7	8.6	12.77	12.8	10.8
		Pole position: 187.8°E, 11.9°S (dp/dm = 6.5/12.5°)				

Dp and dm are the radii of the oval of confidence about the pole position in the colatitude direction and at right angles to it respectively; other symbols are as for Table 3.

[1] Incorporates 19 sites in dykes from Halls et al. (2000), 5 sites from Hou et al. (2001), 2 sites from Zhang and Piper (1994), 1 site from Qian and Chen (1987) and remainder from this study.

[2] Incorporates 4 sites in dykes from Halls et al. (2000), 3 sites from Qian and Chen (1987) and the remainder from this study.

[3] Incorporates 7 sites from Hou et al. (2001), 3 sites from Halls et al. (2000) and the reminder from this study.

[4] Incorporates 1 site each from Halls et al. (2000), Hou et al. (2001) and Zhang and Piper (1994) and the remainder from this study. Pole positions for this study are calculated for a men location at 39°N, 113°E and overall positions for the North China Shield are calculated for a mean site location of 38°N, 113°E.

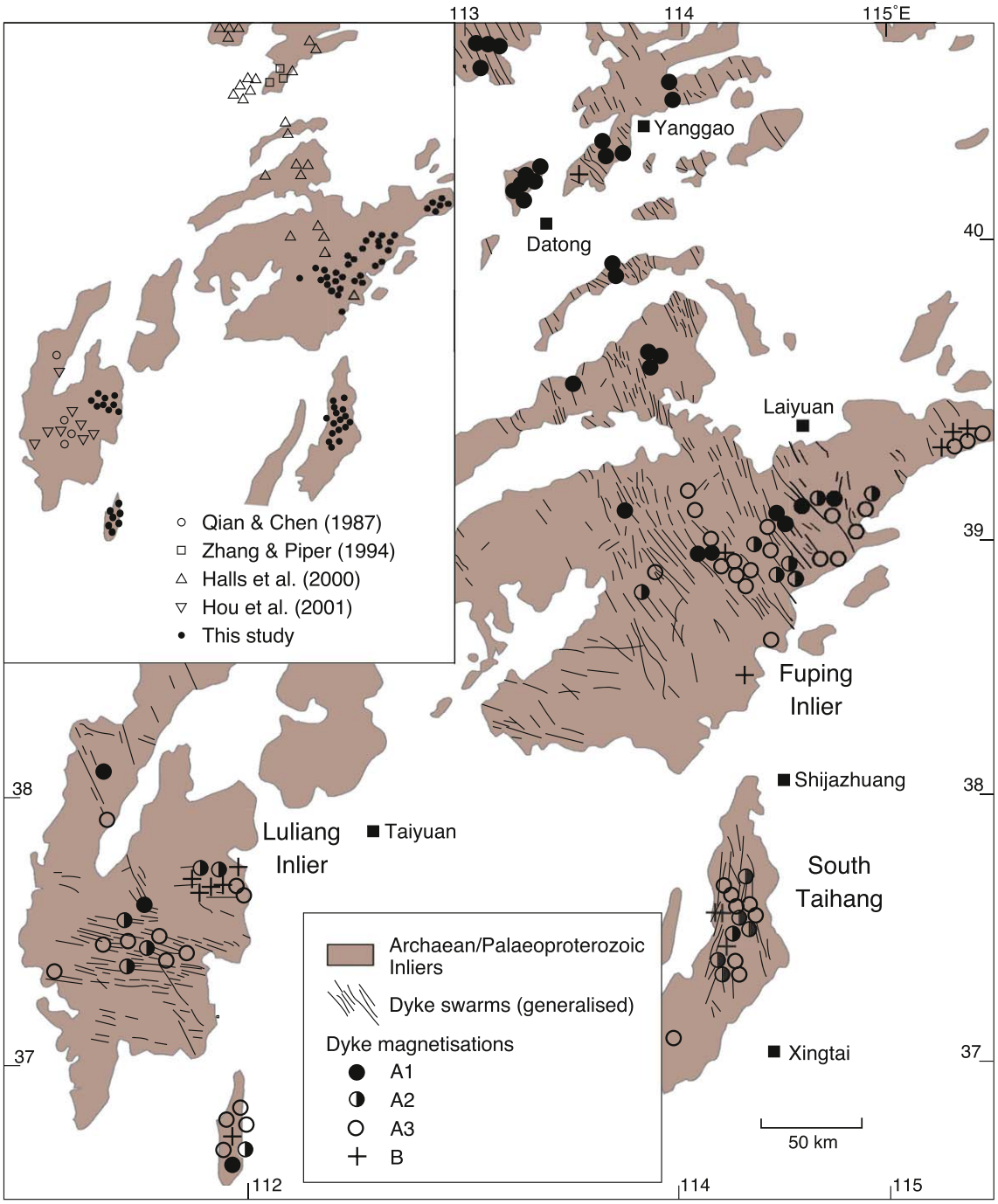


Fig. 14. The areal distribution of A1, A2, A3 and B magnetisations in dykes and country rocks within the Central Zone of the North China Shield. The extent of this map is shown by the dashed box in Fig. 1 and the terrane boundaries are after Peng et al. (2005).

in a marginal oceanic setting by convergence between Yilgarn and Pilbara cratons in Palaeoproterozoic times (Myers, 1990).

The link between Western Australia and Southern Africa is of Archaean antiquity (Zegers et al., 1998) and suggests a wider link between the Capricorn and Limpopo Belts, with the latter finally consolidating in a 2.0–1.9 Ma tectonothermal event recording final suturing of the Zimbabwe and Kaapvaal cratons (Zhao et al., 2002a,b); palaeomagnetic data support inclusion of the Indian Archaean nuclei within this assemblage by early Palaeoproterozoic times (Mondal et al., 2009). The West Africa and Guiana cratons were in broad continuity by ~2.1 Ga as required by widespread development of a fluvio-deltaic sedimentary cover (Rogers, 1996) so that the Transamazonian and Eburnean orogenic belts with ~1.9 Ga syntectonic granites recording the culminating phase of Palaeoproterozoic consolidation are plausibly regarded as components of the same system (Fig. 16). Further south the South America-West Africa link is reinforced by the Francevillian Unit (Gabon) and the Jacobina Unit (Brazil) deposited in foreland basins during collisional orogeny with N–S trend at ~2.0 Ga (Zhao et al., 2002a,b, 2004a,b,c). Although the intervening crust is largely unexposed, the palaeomagnetic link and the trend of these orogens in central-west Africa suggest continuity with contemporaneous orogenic activity in Siberia and western Laurentia (Fig. 16).

Table 4
Palaeomagnetic poles from igneous rocks of the continental shields assigned to the interval ~1.85–1.75 Ga.

	Shield/rock unit	Age	Pole position		dp/dm or A_{95} (°)	Polarity	Code	Rotated pole
			°E	°N				°E/°N
(i) Laurentia								
L1	Wathaman Batholith	1854 ± 11	293	9	3/5	0	6448	–
L2	Macoun Lake Granodioorite	1854 ± 10	288	44	14/15	0	8367	–
L3	Reynard lake Pluton	1851 ± 3	253	40	9/9	0	8361	–
L4	Davina Lake Granodiorite	1850 ± 20	267	54	9/9	20	8367	–
L5	Wekach Lake Gabbro	1849 ± 3	225	1	8/11	0	8362	–
L6	Hanson Lake Pluton	1844 ± 2	266	36	10/10	12	8366	–
L7	Boot Phantom Pluton	1838 ± 1	279	62	8/8	6	8359	–
L8	Deschambault/ Tower Is. plutons	1796 ± 15	258	77	6/6	0	8358	–
L9	Dubawnt Group	1785 ± 4	277	7	8/8	63	2739	–
L10	Deschambault Pegmatites	1770 ± 2	276	68	8/8	0	8889	–
L11	Jan Lake Granite	1767 ± 1	270	48	15/15	0	8371	–
L12	Cleaver Dykes	1740 ± 5	277	19	6/6	0	9139	–
(ii) Fennoscandia								
F1	Hautavaara Gabbro	1880 ± 60	272	34	10/10	0	7428	287/−4
F2	Tsuomasavarri Ultramafic rocks	1850 ± 50	245	26	5/9	100	7525	263/−9
F3	Hankiresi Intrusives	1838 ± 2	225	49	2/4	0	1312	256/18
F4	Keruu Dykes	1870 ± 50	229	42	6/10	27	5739	256/11
F5	Nilsia-Varpaisjarvi Dykes	1845 ± 15	224	47	3/5	0	1327	255/17
F6	Ropruchey Sill	1770 ± 12	230	40	8/8	0	7406	256/8
F7	Luantari Granodiorite	1778 ± 12	198	67	8/10	0	1309	256/41
F8	Tarendo Gabbro	1757 ± 43	230	45	3/5	0	1333	258/13
F9	Tarendo Intrusives	1757 ± 43	237	45	7/11	0	5771	262/11
F10	Post-Svecofennian Intrusions	1780 ± 50	201	48	4/4	0	7535	242/26
(iii) South America (Guyana-Amazonia)								
1	Colider Volcanics	1789 ± 7	299	−63	10	43	Ref. 1	252/11
2	Roraima Dolerites	1500–2090*	231	63	9/9	0	3271	289/41
3	Kabaledo Dykes	~1750*	210	44	10/20	0	3004	271/61
(iv) Africa								
1	Aftout Gabbro	1869 ± 50	55	29	6/8	0	2320	249/37
2	Mashonaland Dolerites	1750–1950	338	8	5/5	62	8097	195/−18
(v) Australia								
1	Plum Tree Volcanics	1822	15	29	–	–	Ref. 2	257/23
2	Hart Dolerite	1762 ± 25	46	29	24/24	83	1940	268/49
3	Lunch Creek Lopolith	1740 ± 30	21	63	9/9	40	1930	293/18
4	Peters Creek Formation	1725 ± 2	41	26	5/5	M	8725	261/47
(vi) North China								
1	Taihang Dykes	1769 ± 3	247	36	2/4	0	8639	283/35
2	A1 Dykes	1769 ± 3	245	38	1/3	10	This paper	284/32
3	A2 Dykes	~1760	281	51	3/7	3	This paper	254/22
4	A3 Dykes	–	241	69	4/7	–	This paper	272/3
(vii) Indian Shield								
1	Dharmapuri dykes	1855 ± 9	260	83	6/11	20	7779	291/50

The pole references are codes in the Global Palaeomagnetic Database.
* Although age constraints are poor for these rocks $^{40}Ar/^{39}Ar$ data suggest an age within the 1.8–1.7 Ga time frame considered here. Ref. 1 is Bispo-Santos et al. (2008). Ref. 2, this result discussed by Schmidt and Williams (2008) and reported in incomplete form by Idnurm and Giddings (1988).

The link between Siberia and Laurentia in Fig. 16 proposed by Sears and Price (1978, 2000) conforms to Mesoproterozoic-Early Neoproterozoic palaeomagnetic data (Piper, 2007) with the Aldan Shield hypothesized to be the source of an Archaean component in the Mojavia terrane near the Laurentian margin; the Akitkan fold belt intruded by ~1.85 Ga syntectonic granitoids (Rosen et al., 1994) is then broadly continuous with the Taltson-Thelon Orogen in Laurentia characterized by 2.0–1.9 Ga granitic magmatism (Sears and Price, 2000). A Palaeoproterozoic Laurentia-Fennoscandia link is well established from palaeomagnetic data (e.g. Patchett et al., 1978; Piper, 1982, 2010b; Buchan et al., 2000) and dismembered by the time of ~1.1 Ga Grenville Orogeny (Piper, 2009); it identifies progressive accretion of subduction-related magmatic arcs bordering the margins of mid-Archaean 'Arctica' nucleus of Rogers (1996) incorporating the Wyoming, Superior, North Atlantic and Karelia cratons (Park, 1992 and Fig. 16). Tectonic evolution of these multiple belts was remarkably concentrated and included collision between the North Atlantic (Nain) and Rae cratons culminating in the Torngat fold belt bordering the Hearne Craton. The latter is in continuity with the Nagssugtoqidian Orogen in Greenland which formed at 1.9–1.8 Ga during convergence of the North Atlantic and Disko cratons (Park, 1995).

The eastward extension of the Torngat and Nagssutoqidian orogens continues through the contemporaneous Laxfordian Belt of NW Scotland to the Kola-Karelia Orogen in Fennoscandia. The sub-parallel Taltson-Thelon Orogen consolidated by convergence of the Slave and Rae Cratons where widespread 2.0–1.9 Ga granitic magmatism has been interpreted as a magmatic arc developed above an east-dipping subduction zone (Hoffman 1988). This is in continuity with the Volhyn-Central Russian Orogen which, although poorly understood, is believed to have consolidated at ~1.9–1.8 Ga by convergence of the Fennoscandia, Volgo-Uralia and Sarmatia nuclei (Park, 1995; Zhao et al., 2004a,b,c). The wider family of 2.0–1.8 Ga orogenic belts, the majority interpreted in terms of continental arc-continental collision processes, includes the Wopmay developed at ~1.95–1.85 by convergence with the

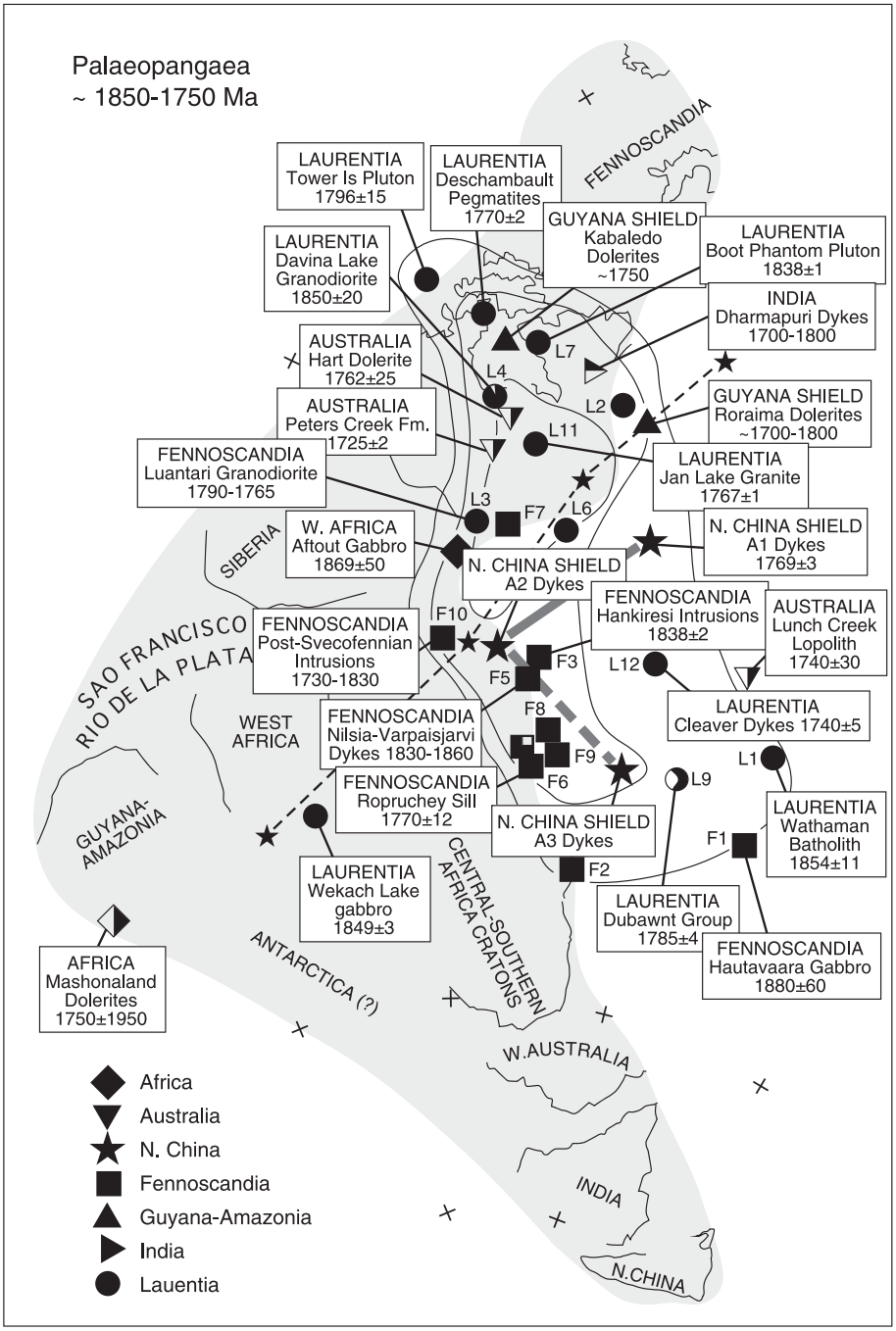


Fig. 15. The Proterozoic supercontinent Palaeopangaea reconstructed after Piper (1982, 2007, 2010a) with minor modification showing rotated palaeomagnetic pole positions from ~1.85–1.75 Ga igneous units listed in Table 4. The closed symbols have a common "normal" polarity (note that affinity with the present day normal field is not clarified in the Palaeoproterozoic) and the open sectors show the proportions with opposite polarity in the palaeomagnetic collections. The NNW-SSE swathe embraced by these igneous poles is comparable to the swathe of palaeopoles, many of uplift-related origin, defined by magnetisations from 1900 to 1700 Ma orogenic belts (Piper, 1982, 1985). Laurentia is retained in present day coordinates and other shields are moved towards it according to Eulerian operations (pole °E, °N, rotation positive when anticlockwise): Africa (138°E, 73°N, −146°), Australia (199.0°E, 23.5°N, +107°), Fennoscandia (8.0°E, 21.0°N, +41.0°), Guyana-Amazonia (246.5°E, −33°N, +142°), India (0.0°E, −7.0°N, −149.5°), North China (265.0°E, 37.0°N, +178.3°), Siberia (217°E, 77°N, +107.3°).

Slave Craton, the Ungava Orogen molded around the core of the Superior Craton, and the Trans-Hudsonian Orogen deformed by convergence of the Superior and Hearne cratons at 1.9–1.8 Ma. Along the southern periphery the Penokean Belt consolidated at 1.9–1.8 Ga by northward emplacement of nappes, whilst the Makkovik Orogen continues via the Ketilidian Orogen in South Greenland through to the Svecofennian Orogen in Fennoscandia (Park 1995; Rogers and Santosh 2004, Shao et al., 2005). These are all apparently accretionary orogens formed at 2.0–1.8 Ga and plausibly linked to a large subduction-related magmatic arc system bordering the 'Arctica' nucleus (Fig. 16) with a close temporal, and possibly genetic, link to the 1.8 Ga LIP event.

8. Conclusions

The post-tectonic dyke swarm trending mostly N-NNW and emplaced into the NCS at ~1.78–1.76 Ga was magnetised in a

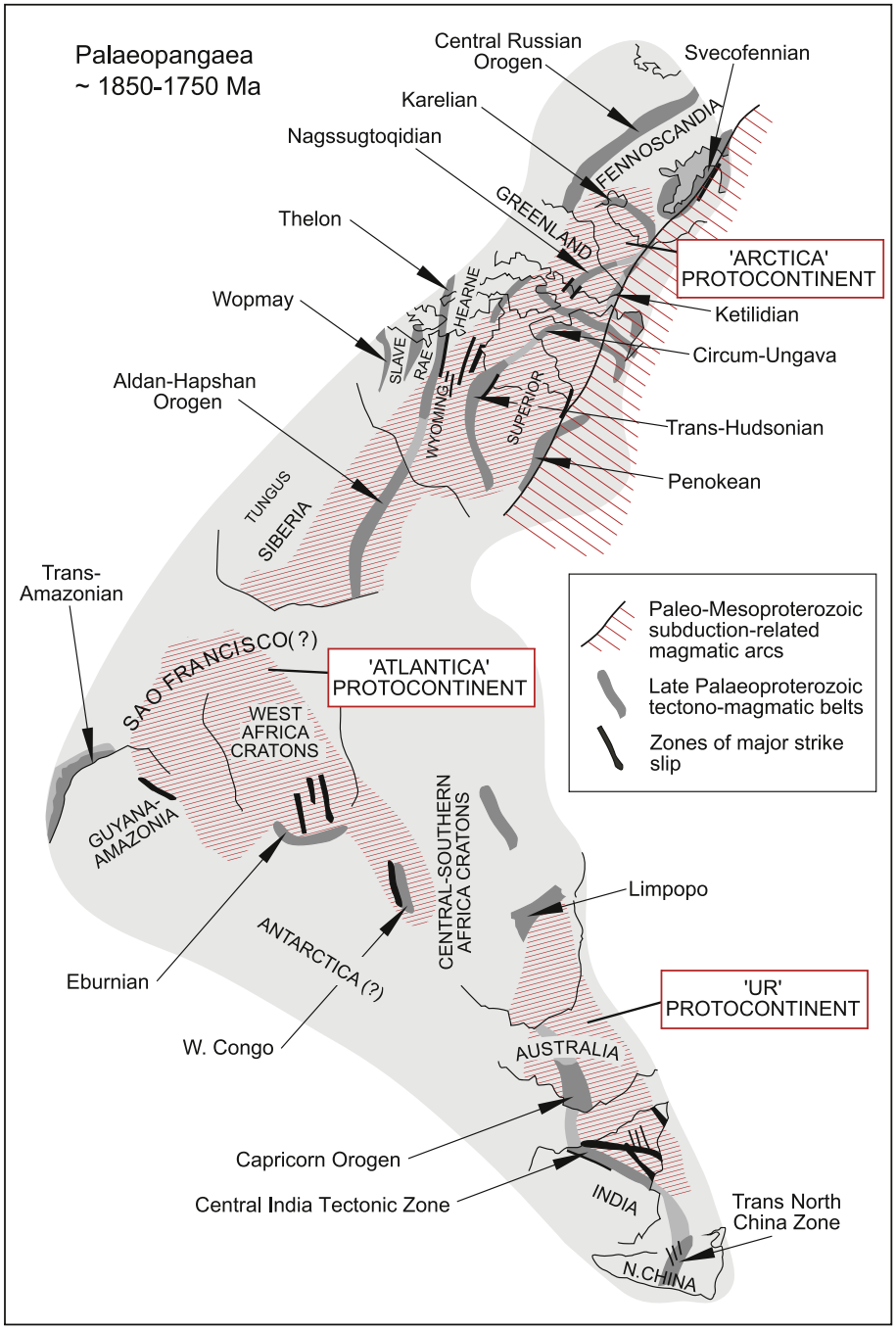


Fig. 16. Palaeopangaea in Palaeo-Mesoproterozoic times showing the distribution of 1900–1700 Ma orogenic belts that consolidated around an Archaean core comprising the 'Ur', 'Arctica' and 'Atlantica' protocontinents (Rogers, 1996; Rogers and Santosh, 2004; Piper, 2010a). Note that the configurations of these continental nuclei were retained with peripheral modification into Neoproterozoic times and finally dismembered by continental break-up shortly after 600 Ma (Piper 2007, 2010b).

geomagnetic field with near-equatorial inclination and a declination that migrated through a ~40° arc embracing a northerly-directed A2 to a NE-directed A1 group; reversals are observed but a single polarity dominated most of the interval of magnetisation. Contact tests indicate a primary origin for this remanence although cooling appears to have occurred during a protracted period of uplift following the culmination of regional granulite facies metamorphism at ~1.85 Ga; this period of extension embraced widespread dyke injection and initiation of aulacogens within the NCS. A subsidiary (B) dyke swarm of dual polarity and E–W directed magnetisation also present in this shield is presently undated but probably of younger Precambrian age. The A1–A2 dykes in common with the basement show increasing degrees of alteration and concomitant remagnetisation as the grade of regional metamorphism declines from granulite facies in the north to amphibolite facies towards the south, and also towards the perimeters of the Precambrian inliers influenced by Mesozoic tectonic and igneous activity. The probable influence of later events is embraced by a third grouping of components (A3) apparently of mixed origin and unqualified age.

The A1 and A2 pole positions group together with magnetisations in host basement to define an APW swathe for the NCS which can be correlated with igneous poles of comparable age from other shields, including results from the ~1.8 Ga LIP events, to evaluate

the Palaeoproterozoic setting of North China. A location bordering the Indian shield is implied and demonstrates continuity of key Palaeoproterozoic geological features. Wider correlation also indicates that these blocks were sited near the periphery of the symmetrical supercontinent Palaeopangaea with tectono-thermal mobile belts accreting to three protoliths, two of which ('Ur' and 'Arctica') had already consolidated by late Archaean times. The Late Palaeoproterozoic tectono-magmatic events in the NCS and Indian shields were part of a wider interval of intense activity at ~1.9–1.7 Ga which included accretion of orogenic belts around this core nucleus and LIP events. The distribution of ~1.78 Ga dykes in the NCS and probable continuation of the swarm into the Indian Shield imply great lateral continuity comparable to the later Mackenzie dyke swarm (1.267 Ga). The observation that these have not been the sites of later continental break-up and the long term integrity of continental crust before and after these LIP-related events inferred from the palaeomagnetic data implies that these are no longer to be correlated with the dismemberment of supercontinents.

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